High-Purity Ti

Common Name: Iodide or electrolytic Ti
UNS Number: Unassigned

High-purity titanium has one-half the oxygen content as commercially pure (ASTM grade 1) titanium. High-purity titanium is produced from a special grade of titanium sponge (<0.1 wt% oxygen).

Chemistry. High-purity titanium generally has oxygen contents of about 500 ppm or less. Electrolytic methods are being used to produce a very hig- purity titanium sponge on a pilot-plant scale.

Product Forms. Pure titanium is supplied as single crystals, crystal bars, and polycrystalline wrought forms.

Density. α phase: 4.51 g/cm^3 (0.163 lb/in.^{3}) at 20 °C. β phase: 4.35 g/cm^3 at 885 °C (1625 °F) from indirect measurements (0.157 lb/in.^{3})

Product Conditions. Cold worked, stress relieved, annealed

Applications. Typical uses include experimentation and research, commercial applications requiring minimum interstitial alloying elements (oxygen, nitrogen, carbon, and hydrogen).

Specifications and Compositions. (see table)

High-purity Ti compositions

Specification designation	Form(s)	C	H	N	O	Cu	Fe	Mn	Sn	Si	Zr	Cl	Mg	Ti bal
Typical %, electrolytic Ti	...	0.008	...	0.004	0.037	0.007	0.009	<0.001	<0.020	0.002	< .001	0.073	<0.001	99.837
Typical %, Iodide Ti	Crystal bar	0.001	...	0.002	0.03 0.06	< 0.001	0.002	0.003	0.001	0.005	0.050	0.002	0.003	99.87
IMI 110, max wt%	Sheet	0.02	0.004	0.005	0.05	0.02	0.025	0.05	0.02	0.02	...	...	...	...

Summary of Physical Properties

Atomic Properties

Atomic Number: 22

Atomic Weight: 47.88 on carbon scale (47.88 g/mole)

Electron Structure: $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$, $3d^2$, $4s^2$

Ionization potentials

Ion	Outer electrons	Ionization potential,V
Ti^+	3d24s	6.83
Ti^{++}	$3d^2$	13.63
Ti^{+++}	3d	28.14

Electrical/ Magnetic Properties

Electrical/ Electronic Properties	
Electrical resistivity(a)	
High purity	42 μΩ · cm
Commercial purity	55 μΩ · cm
Superconductivity, critical temperature	0.37 to 0.56 K
Hall coefficient	$+1.82 \pm 0.2 \times 10^{-13}$ volt · cm/amp/oersted
Work function	4.17 eV
Band width (from soft X-ray spectra)	6.0 ± 0.5 eV
Electronic specific heat	8.00×10^{-4} cal/°C · mole

(a) Room-temperature propertites iodide titanium or relatively pure commerical titanium with a minimum yield strength of 275 to 380 mPa (40 to 55 ksi). Source: Lepkowski and Holladay, TML ReportNo. 73, Battelle, 1957. See subsequent sections of this datasheet for additional data on electrical resistivity and other physical properties.

Magnetic Properties	
Magneto-resistance coefficient	6.6×10^{-13}/oersted^2
Magnetic susceptibility	$3.17 \times 10^{-6}(\pm 0.03 \times 10^{-6})$emμgram
Relative magnetic permeability	1.00005

Source: Lepkowski and Holladay, TML Report No. 73, Battelle, 1957

Mass Properties

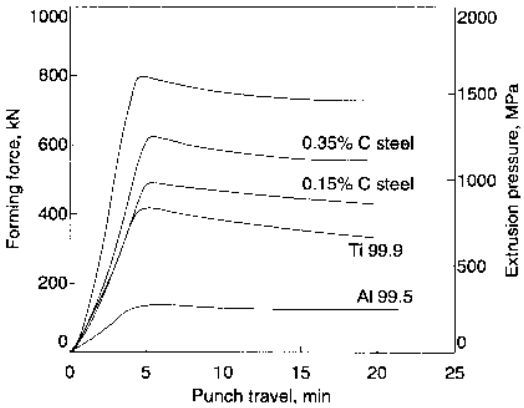
Compressibility. 0.80×10^{-6} cm^2/kgf at room temperature

Density. α phase, 4.507 g/cm^3 at 20 °C
β phase, 4.35 g/cm^3 at 885 °C

Lattice parameters of α: c = 0.468 nm; a = 0.295 nm

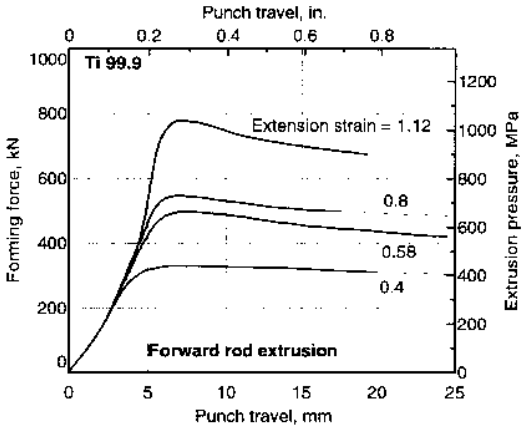
Lattice parameters of β: a = 0.332 nm at 900 °C

Pure titanium: Extrusion pressure vs other materials



Forward rod extrusion with an extrusion strain of 0.85. Composition (wt%) of pure titanium was 99.9 Ti, 0.04 Fe, 0.04 O_2, 0.01 N_2, 0.002 H_2, and 0.010 C.
Source: H.W. Wagener and K.H. Tampe, Cold Impact Extrusion of Titanium, *Titanium Science and Technology*, Vol 1, DGM, 1985, p 577-584

Pure titanium: Force-travel diagrams of cold extrusion

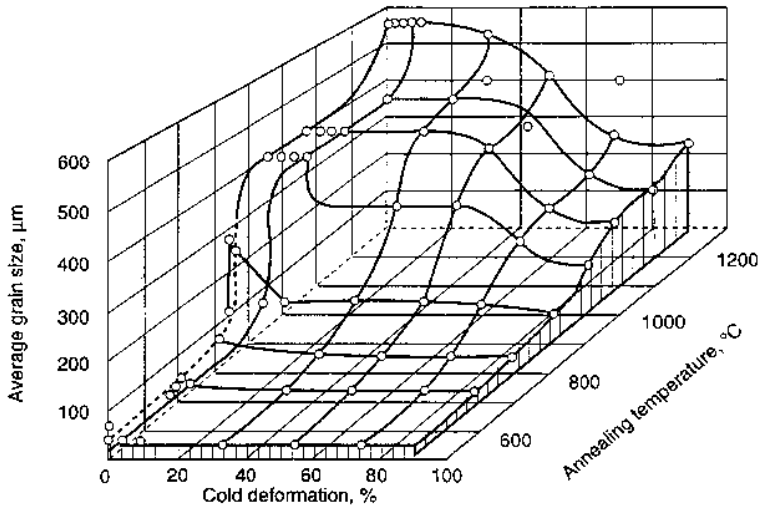


99.9% Ti with 0.04% O_2, 0.01% N_2, 0.002% H_2, 0.04% Fe, and 0.010% C
Source: H.W. Wagener and K.H. Tampe, Cold Impact Extrusion of Titanium, Titanium Science and Technology, Vol 1, DGM, 1985, p 577-584

Heat Treatment

In keeping with commonly observed behavior, increasing amount of cold work prior to annealing decreases the subsequent recrystallization temperature and grain size (see figure). Increasing the amount of cold work before annealing also decreases the spread in grain volume distribution (*Met. Trans.*, Vol 16A, May 1985, p 703-708). The reason for the decrease in grain size distribution is not completely clear.

Iodide Ti: Grain size vs annealing and deformation



Influence of annealing temperature (1-h duration) and degree of deformation on the grain size of cold-deformed sheet-strips of iodide titanium.
Source: U. Zwicker, *Titan und Titanlegierungen*, Springer-Verlag, 1974, p 138

Hall-Petch Parameters

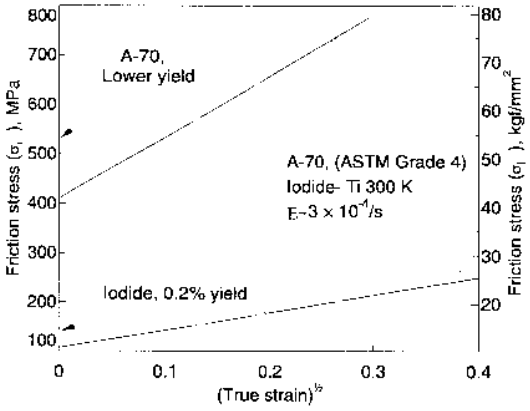
Interstitial impurities increase both the Hall-Petch friction stress (σ_f) and the Hall-Petch grain size constant (K) at room temperature (see comparisons with ASTM Grade 4). As temperatures increase, interstitial impurities have less of an effect (see figure).

Effect of impurities on Hall-Petch parameters at room temperature

Material/ composition(a)	Hall-Petch parameters(b) Friction stress (σ_f) MPa	kgf/mm^2	Grain size constant (K) kgf/mm$^{3/2}$
Iodide titanium (30 O, 18 C, 110 N)	137	14	0.47
Battelle titanium (330 O, 25 C, 140 N)	176	18	0.32
A-70 (ASTM grade 4)	390	40	0.86

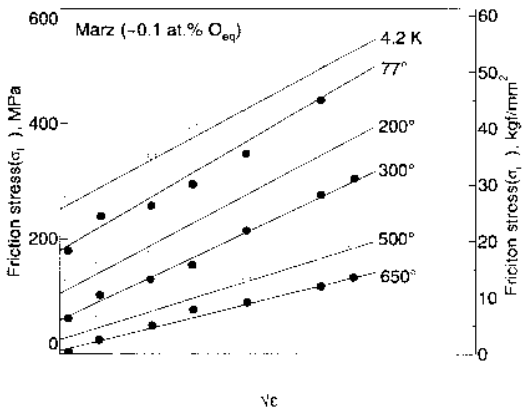
(A) All compositions in ppm by weight. (b) Hall-Petch friction stress (σ_f) and grain size constant (K) used to model flow stress (σ) by the relation $\sigma = \sigma_i + K\sqrt{d}$, where d is the grain size. Source: H. Conrad and R. Jones, *The Science, Technology and Application of Titanium*, Jaffee and Promisel, Ed., Pergamon Press, 1970, p 495

Hall-Petch friction stress of pure titanium and ASTM Grade 4

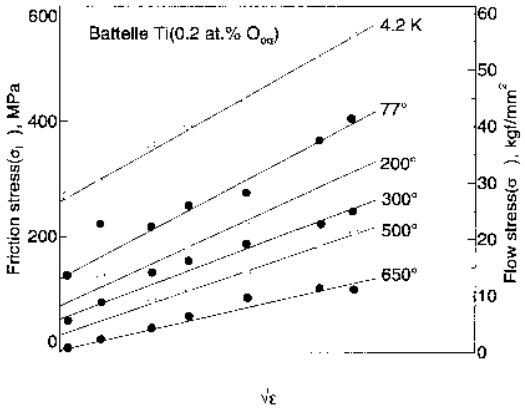


Room temperature data for Grade 4 and iodide titanium (30 ppm O, 110 ppm N) for various strains.
Source: R.L. Jones and H. Conrad, *Trans. TMS/AIME*, Vol 245, 1969, p 779-789

Effect of strain and temperature on friction stress



(a)



(b)

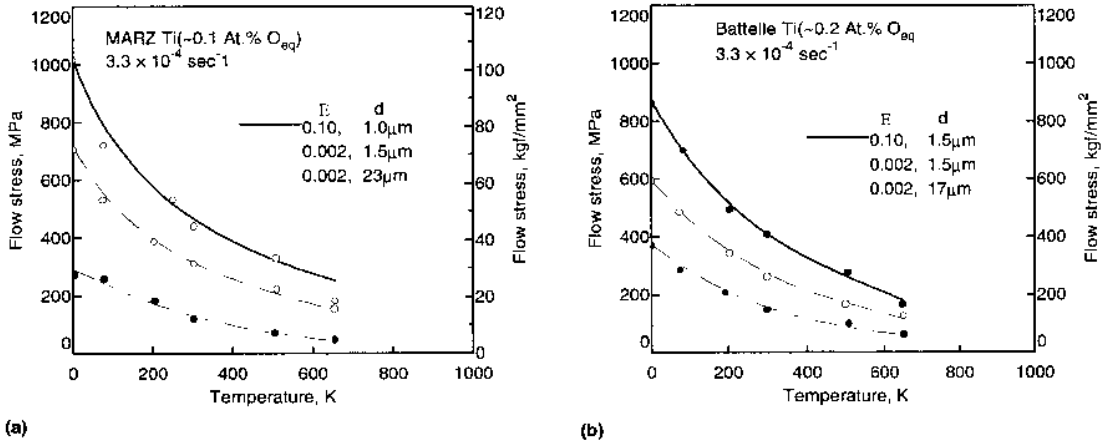
Strain rates were 3.3×10^{-4}/s.
Source: K. Okazaki and H. Conrad, *Acta Metall.*, Vol 21, 1973, p 1117-1129

Cold-Impact Extrusion

The main characteristics of extrusion can be described with force-travel diagrams (see figure). Cold-impact extrusion allows close dimensional control and the advantage of work hardening.

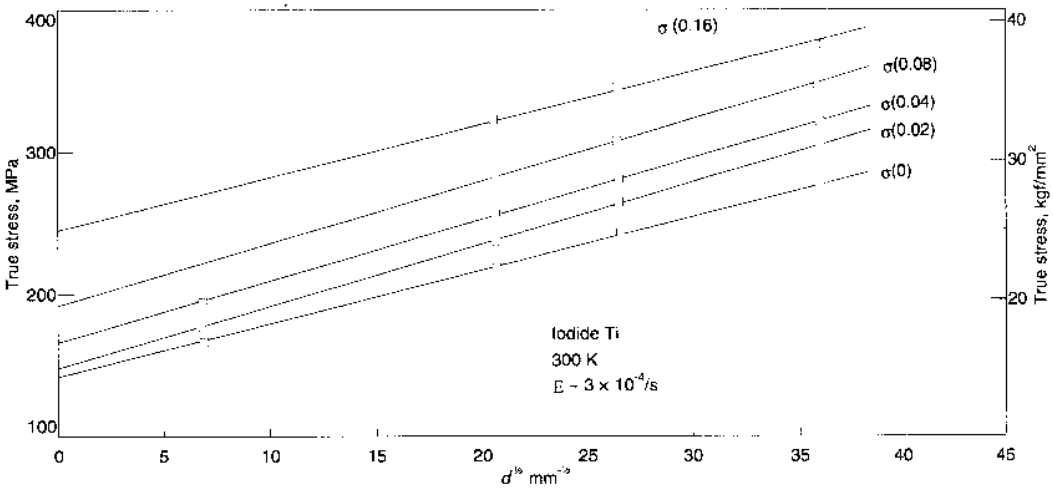
Effect of Grain Size

Pure titanium: Effect of temperature on flow stress



(a) MARZ titanium with 182 to 790 ppm O by weight and 0.11 at.% O_{eq}. (b) Battelle titanium with 330 to 569 ppm O by weight and 0.18 at.% O_{eq}. Both compositions tested at a strain rate of 3.3×10^{-4}/s.
Source: K. Okazaki and H. Conrad, *Acta Metall.*, Vol 21, 1973, p 1120

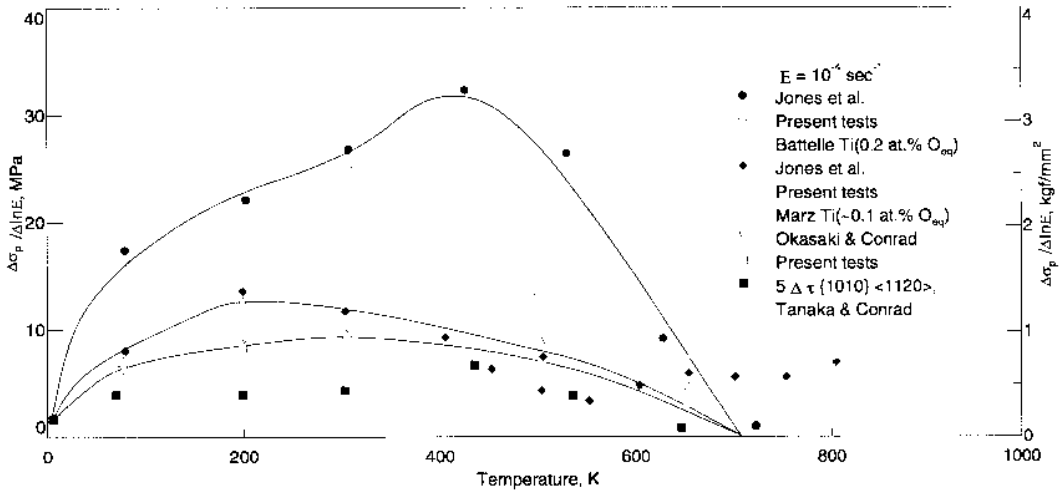
Effect of grain size (*d*) on flow stress



Room-temperature flow stress of iodide titanium (30 pm O, 110 pm N) for plastic strains of 0, 0.02, 0.04, 0.08 and 0.16.
Source: R.L. Jones and H. Conrad, *Trans. TMS/AIME*, Vol 245, 1969, p 779-789

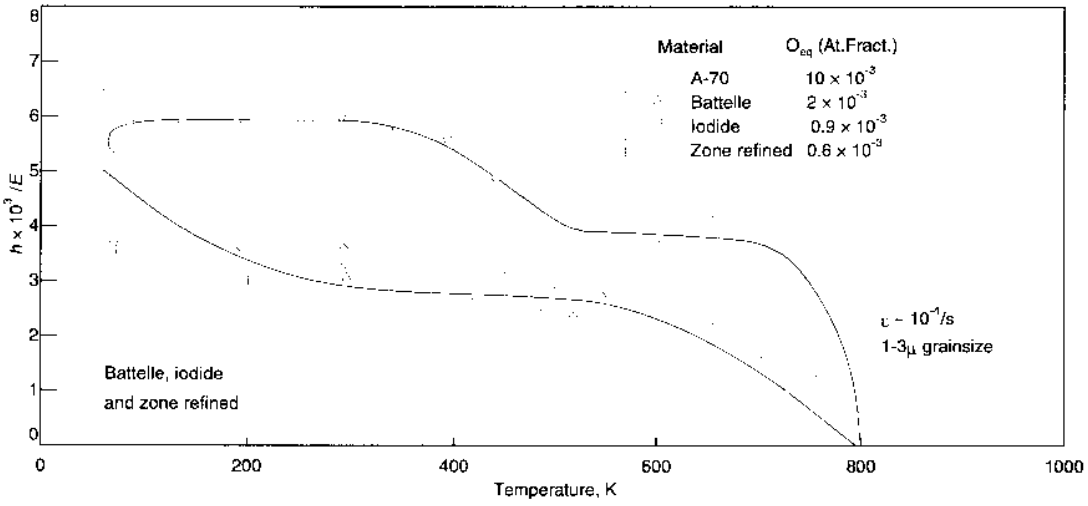
Strain-Rate Sensitivity

Effect of impurities on strain rate sensitivity



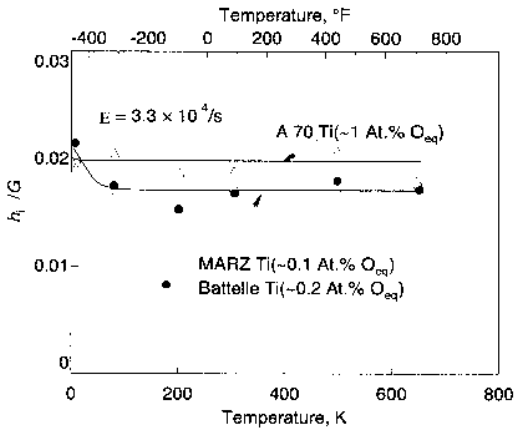
Source: K. Okazaki and H. Conrad, *Acta Metall.*, Vol 21, 1973, p 1122

Pure titanium: Temperature effect on strain hardening coefficient (*h*)



Normalized to the modulus of elasticity (*E*).
Source: H. Conrad and R. Jones, *The Science, Technology and Application of Titanium*, Jaffee and Promisel, Ed., Pergamon Press, 1970, p 492

Pure titanium: Effect of temperature on h_l/G



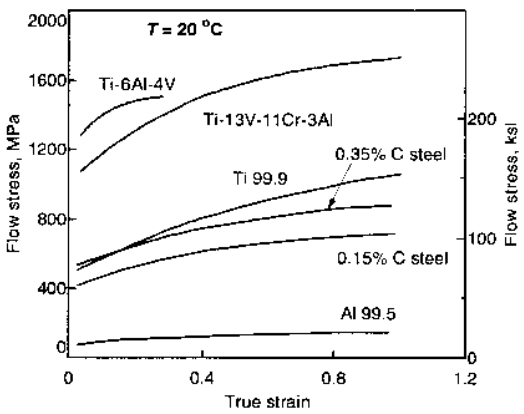
The strain hardening constant (h_l) is normalized with the shear modulus (*G*), where *G* is the C_{66} constant (taken from E.S. Fisher and C.J. Renkin, *Phys. Rev.*, Vol 135, 1964, p A482).
Source: K. Okasaki and H. Conrad, *Acta Metall.*, Vol 21, 1973, p 1121

Flow Stress

Pure titanium (99.9% Ti) has a flow stress comparable to medium-carbon (0.35% C) steel (see figures) and can be cold forged or formed. However, due to the tendency of titanium to stick, fret, and cold weld with steel tooling, cold working of titanium generally requires surface treatment and/or lubrication.

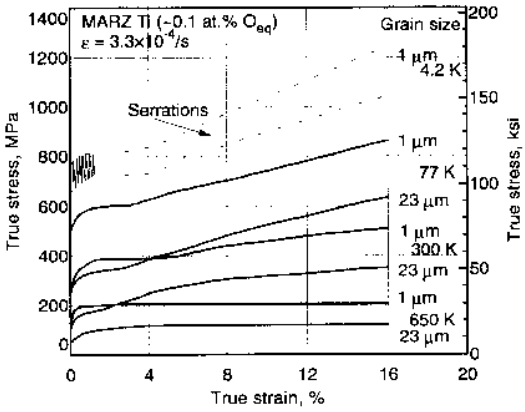
Surface Treatment. The sticking of titanium with steel tools during forming can be overcome with lubricants such as molybdenum disulfide. Physical vapor depositions (such as ion-plated copper) also provide lubrication during cold-working operations. For hot-working operations, lubricants include yttria ceramic coatings.

Pure titanium: Flow stress vs other materials



Titanium impurities (wt%) were 0.04 O_2, 0.01 N_2, 0.002 H_2, 0.04 Fe, and 0.010 C.
Source: H.W. Wagener and K.H. Tampe, Cold Impact Extrusion of Titanium, *Titanium Science and Technology*, Vol 1, DGM, 1985, p 578

Stress-strain curves of zone-refined titanium



MARZ titanium, ~0.1 at.% O_{eq}, with indicated grain sizes (Gs). Source: K. Okazaki and H. Conrad, *Acta Metall.*, Vol 21, 1973, p 1119

Strain Hardening

The strain hardening relation $\sigma = \sigma_0 + h\sqrt{\varepsilon}$ is affected by interstitial content, as shown by a comparison of σ_0 and h for iodide titanium and ASTM Grade 4 (see table). The effect of grain size on h for pure titanium is also less than that of ASTM Grade 4 (see table). Temperatures between room temperature and 325 °C (600 K) have little effect on the ratio of h with selected elastic constants (see figures).

Comparison of strain hardening parameters

Material	Grain size, µm	σ(0)(a) MPa	σ(0)(a) kgf/mm²	Strain hardening coefficient, h MPa	Strain hardening coefficient, h kgf/mm²
ASTM Grade 4 (A-70)	18 to 19	510	52	833	85
ASTM Grade 4 (A-70)	2.6	570	58	705	72
ASTM Grade 4 (A-70)	0.8	715	73	600	61
Iodide (30 ppm O, 110 ppm N)	3.3	215	22	343	35
Iodide (30 ppm O, 110 ppm N)	1.1	265	27	285	29

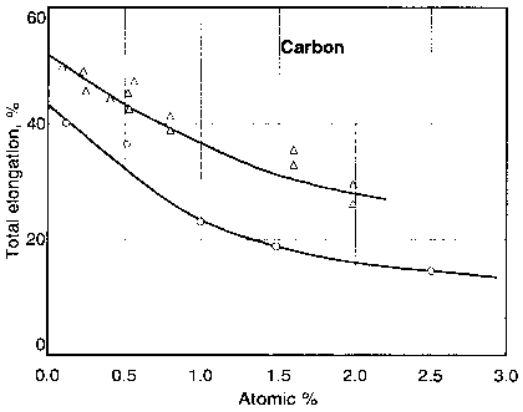
(a) σ(0) = the maximum stress for zero plastic strain. Source: R.L. Jones and H. Conrad, Transactions of the Metallurgical Society of AIME, Vol 245, 1969, p 779-789

Pure titanium: Effect of grain size on the strain hardening coefficient (*h*)

Material/ composition(a)	Grain size parameters(b): h_i MPa	h_i kg/mm²	K' kg/mm$^{3/2}$
Iodide titanium (30 O, 18 C, 110 N)	440	45	0.5
Battelle titanium (330 O, 25 C, 140 N)	353	36	0.2
A-70 (ASTM grade 4)	872	89	0.8

(A) All compositions in ppm by weight. (b) Grain size effects on h modeled by the equation $h = h_i + K'\sqrt{d}$ where d is the average grain size. Source: H. Conrad and R. Jones, *The Science, Technology and Application of Titanium*, Jaffee and Promisel, Ed., Pergamon Press, 1970, p 495

Effect of carbon on elongation at fracture (300K)



(c)

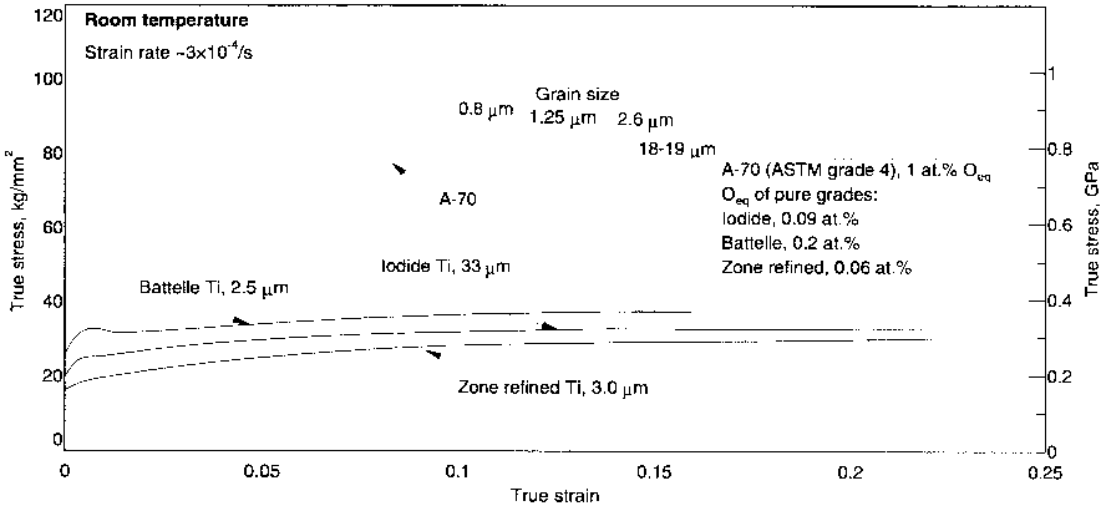
Source: H. Conrad *et al.*, *Titanium Science and Technology*, Plenum Press, 1973, p 997

Plastic Deformation

Stress-strain plots from two different sources (see figures) indicate data scatter similar to that of ASTM grade 4 (A-70) titanium. Typically, the stress-strain behavior after yielding obeys a parabolic relation such that

$\sigma = \sigma(0) + h\sqrt{\varepsilon}$, where σ is the flow stress, $\sigma(0)$ is the intercept, and h is the strain hardening coefficient. However, a linear strain hardening relation has been observed for coarser (28 μm) grained specimens at larger (8%) strains (H. Conrad and R. Jones, *The Science, Technology and Application of Titanium*, Jaffee and Promisel, Ed., Pergamon Press, 1970, p 493).

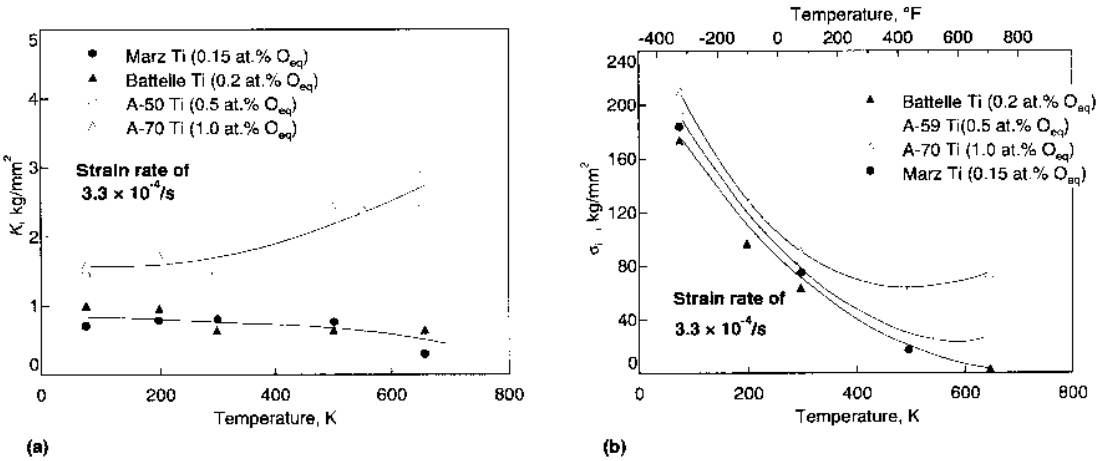
Stress-strain curves of four purities of titanium



Oxygen equivalents (O_{eq}) calculated according to: N = 2×O and C = 3/4×O, in at.%. Analysis by weight in ppm: Zone refined (<30 O, 18 C, 110 N, 18 H); Iodide (30 O, 18 C, 110 N, 18 H); Battelle (330 O, 25 C, 140 N, 62 H, 90 Fe).

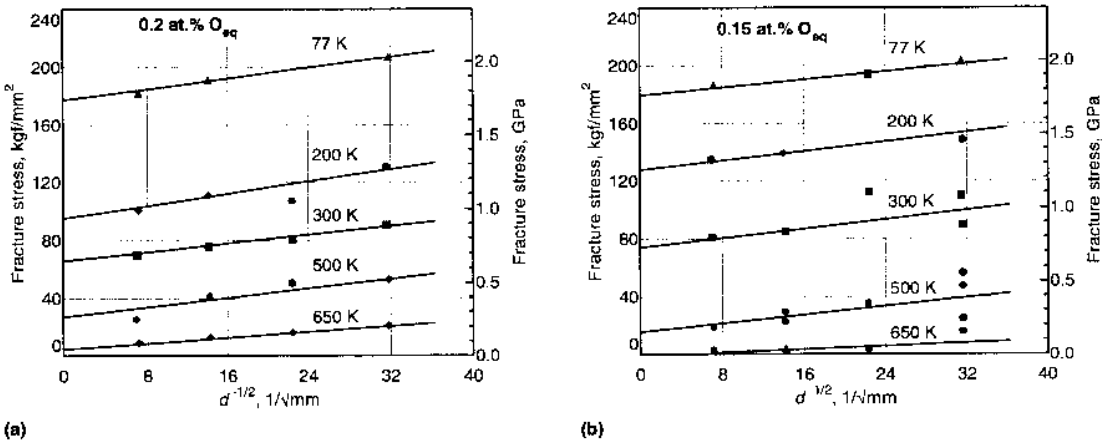
Source: H. Conrad and R. Jones, Effects of Interstitial Content and Grain Size on the Mechanical Behavior of Alpha Titanium Below 0.4 T_m. *The Science, Technology and Application of Titanium* (Jaffee and Promise, Ed.), Pergamon Press, 1970, p 491

Pure titanium: Hall-Petch coefficients for fracture stress (σ_F)



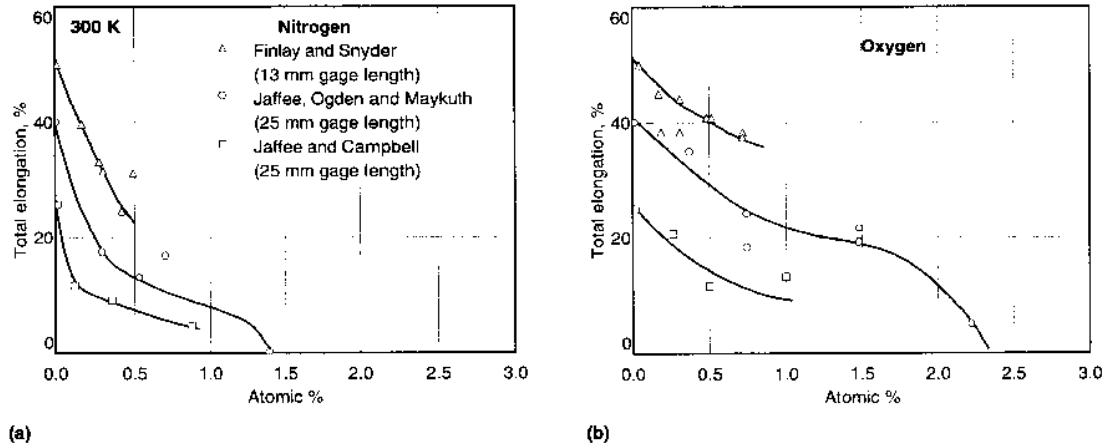
The results take into account the Bridgman correction factor for necking and show the effect of interstitial content on (a) the grain size constant (K_F) and (b) the friction stress (σ_i). $\sigma_F = \sigma_i + K_F\sqrt{d}$ where d is the average grain size.
Source: G. Sargent *et al.*, Effect of Temperature, Interstitial Content and Grain Size on the Fracture Behavior of Titanium, *Titanium and Titanium Alloys* (Williams and Belov, Ed.), Plenum Press, 1982, p 679-689

Pure titanium: Effect of grain size and interstitial content on fracture stress



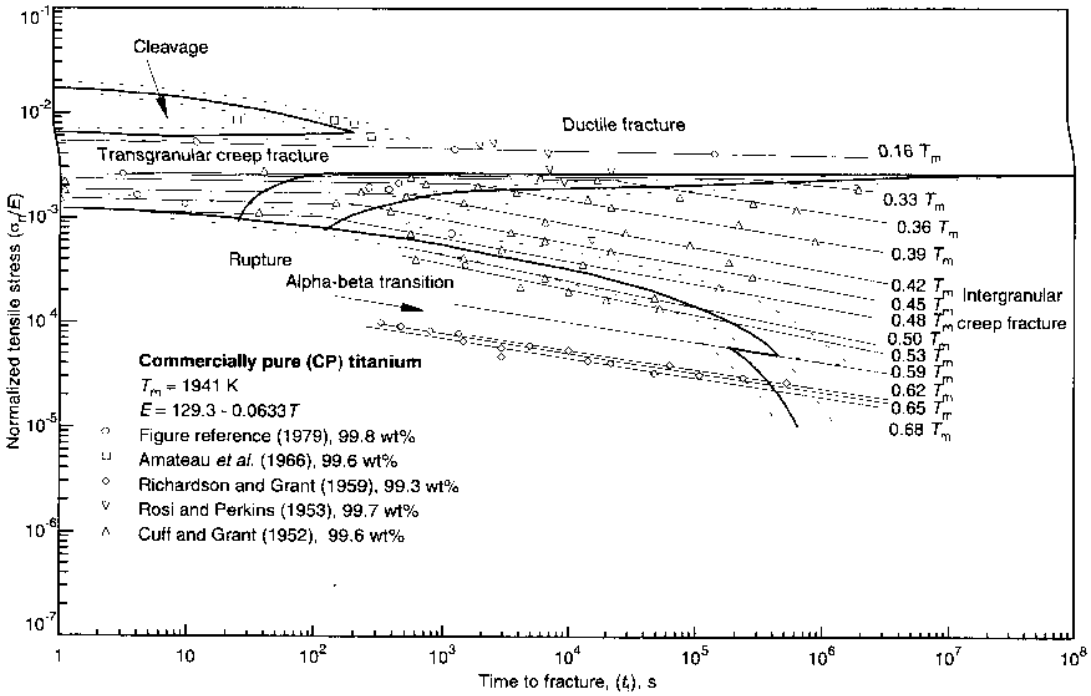
Strain rate was 3.3×10^{-4}/s.
Source: G. Sargent *et al.*, Effect of Temperature, Interstitial Content and Grain Size on the Fracture Behavior of Titanium, *Titanium and Titanium Alloys*, Williams and Belov, Ed., Plenum Press, 1982, p 684

Effect of N, O, and C interstitials on elongation at fracture (300K)



Source: H. Conrad *et al.*, *Titanium Science and Technology*, Plenum Press, 1973, p 997

CP titanium: Fracture mechanism map



Full citations of data sources are given in the figure reference. The region of intergranular creep fracture is based on the data of Cuff and Grant (*Iron Age*, Vol 20, p 134), which attributes a sharp break in in stress-rupture plots to change in fracture mode (i.e. transgranular to intergranular). Source: P. Rama Rao *et al.*, *Acta Metall.*, Vol 34, 1986, p 1783-1806

Fracture Stress

The fracture stress (σ_F) below 0.3 T_m obeys a modified Hall-Petch relation:

$$\sigma_F = \sigma_F + K_F\sqrt{d}$$

where σ_F is the friction stress, K_F is the grain size constant, and d is the average grain size. The value of K_F increases with interstitial content, especially above 500 K (400 °F) where dynamic strain aging becomes more pronounced with higher interstitial contents (see figure). The friction stress (σ_i) is a function of interstitial content, strain rate, temperature, and the amount of strain.

Elongation at fracture decreases with decreasing interstitial content but is relatively independent of grain size (H. Conrad *et al.*, Critical Review Deformation and Fracture, *Titanium Science and Technology*, Vol 2, Plenum Press, 1973, p 996).

Results of impact and slow bend tests on Ti-H alloys

Hydrogen content, Atomic Pct	Energy absorbed in breaking, ft-lb: Impact tests 18.1 ft. per sec	Slow bend tests, 0.08 in. per Min
0.05	30	15.5
0.05	32	20
0.26	19.5	18.5
0.26	20	22
2.0	3.5	4.0
2.0		6.5

Source: G.A. Lenning, C.M. Craighead, and R.J. Jaffee, Constitution and Mechanical Properties of Titanium-Hydrogen Alloys, *Hydrogen Damage*. (C.D. Beachem, Ed.), American Society for Metals, 1977, p 100

Fracture Mechanisms

Pure (99.9%) titanium has only three fracture modes (see figure), all of which are ductile modes of fracture. Brittle fracture is absent at low temperatures and at high temperature-low stress conditions.

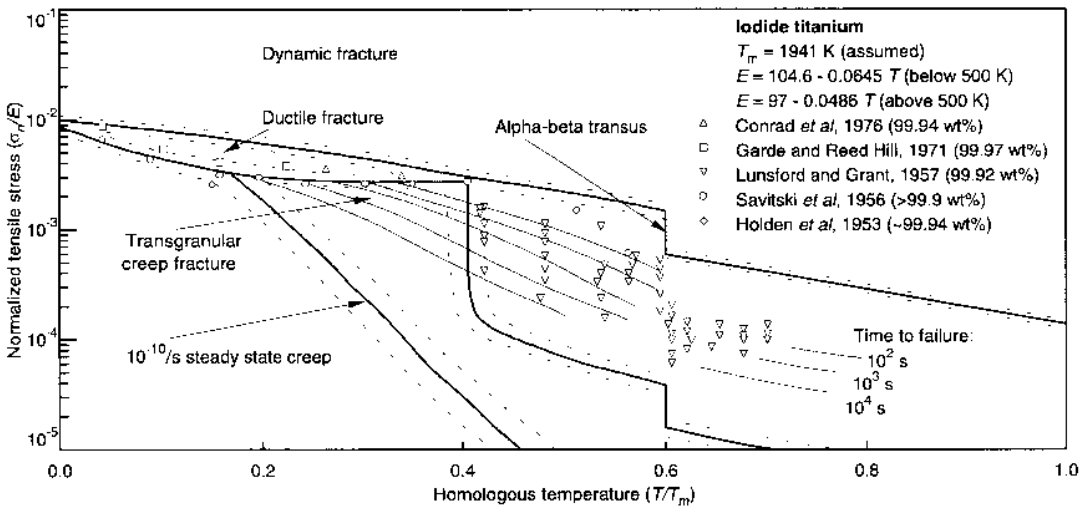
Below 0.3 T_m fracture stress obeys the Hall-Petch relation, while fracture above 0.4 T_m correlates with the following relation for stress rupture:

$$\sigma_r = f[\, t_r \exp(-H/RT)\,]$$

where σ_r is the rupture stress, t_r is the time to rupture, and H is a constant (which is considered to be the activation energy for creep and self-diffusion in the case of pure metals). This relation developed by Orr *et al.* (*Trans Amer. Soc. Metals*, Vol 46, 1954, p 113) identifies an activation energy of 250 kJ/mole (60 kcal/mole) for commercially pure (99.6% Ti) grade 75A. This activation energy is considered a close estimate for the activation energy of pure titanium.

Effect of Impurities. The reduction of ductility with high interstitial contents distinguishes the fracture behavior of high purity titanium (99.9%) from that of commercially pure grades. Rosi and Perkins (*Trans. Amer. Soc. Metals*, Vol 45, 1953, p 972), for example, observed a brittle (cleavage mode) failure of 99.7% Ti specimen [(25 μm grain size at 77 K (−195 °C)]. A recent review of other fracture tests also identifies a region of brittle fracture for commercially pure (mostly 99.7 wt%) titanium (see fracture mechanism map).

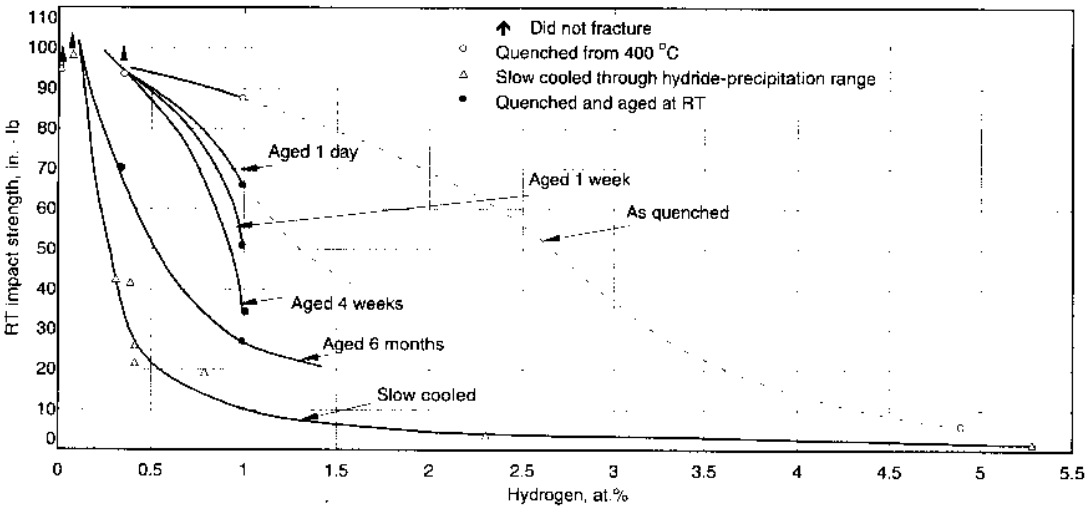
Iodide titanium: Fracture mechanism map



The left boundary for transgranular creep fracture is chosen at a steady state creep rate of $10^{-10}\,s^{-1}$, per Gandhi and Ashby, *Acta Metall.*, Vol 27, 1979, p 1565. Parameters normalized with Young's modulus (E) and melting temperature (T_m).
Source: P. Rama Rao *et al.*, Fracture Mechanism Maps for Titanium and its Alloys, *Acta Metall.*, Vol 34, 1986, p 1783-1806

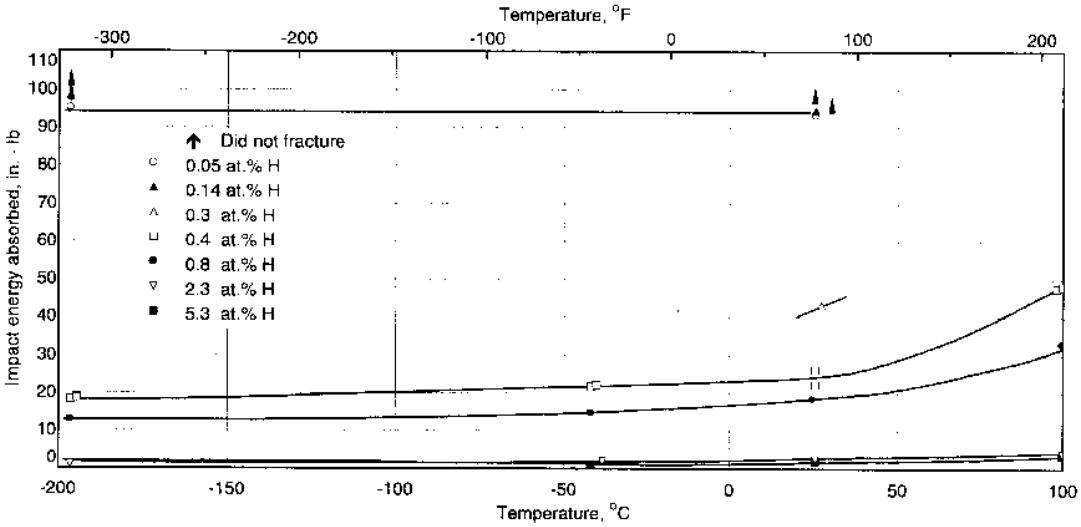
Impact Strength

High-purity Ti: Effect of hydrogen and heat treatment on impact strength



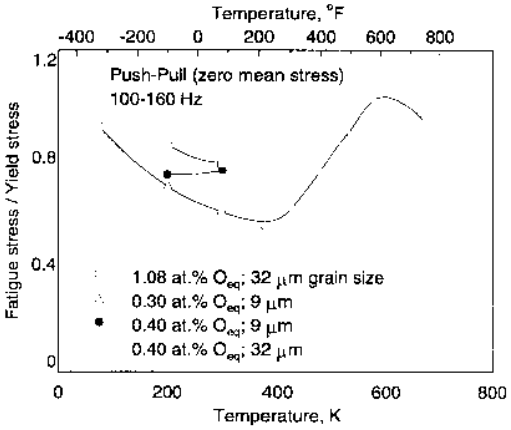
The beneficial effect on toughness of water quenching from α phase field occurs from the fine dispersion of hydrides formed during precipitation. Slow cooling precipitates plate-like hydrides.
Source: G.A. Lenning, C.M. Craighead, and R.I. Jaffee, Constitution and Mechanical Properties of Titanium-Hydrogen Alloys, *Hydrogen Damage*(C.D. Beachem, Ed.), American Society for Metals, 1977, p 100

High-purity Ti: Effect of hydrogen on impact strength



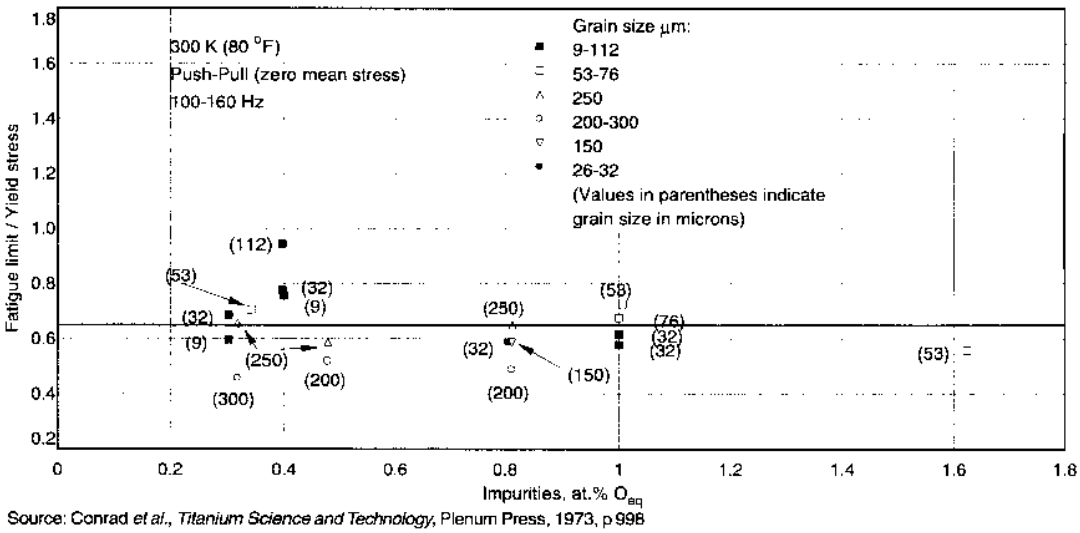
Effect of hydrogen on impact properties of high purity titanium. Specimens cooled through hydride-precipitation range at about 15 to 20 °C per min.
Source: G.A. Lenning, C.M. Craighead, and R.I. Jaffee, Constitution and Mechanical Properties of Titanium-Hydrogen Alloys, *Hydrogen Damage* (C.D. Beachem, Ed.), American Society for Metals, 1977, p 100

Pure Ti: Yield stress/fatigue limit ratio vs temperature



Source: N.G. Turner and W.T. Roberts, *J. Less-Common Met.*, Vol 16, 1968, p 37; and Trans. AIME, Vol 242, 1968, p 1223

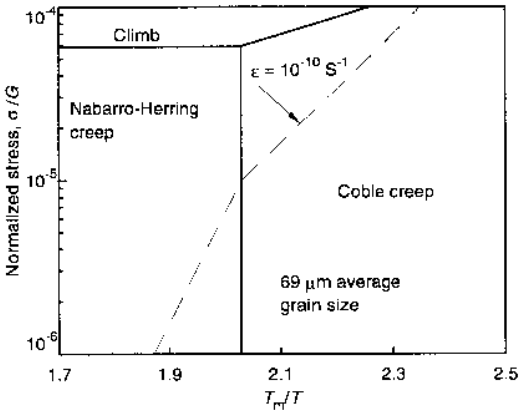
Unalloyed titanium: Ratio of yield strength and fatigue limit



Source: Conrad *et al.*, *Titanium Science and Technology*, Plenum Press, 1973, p 998

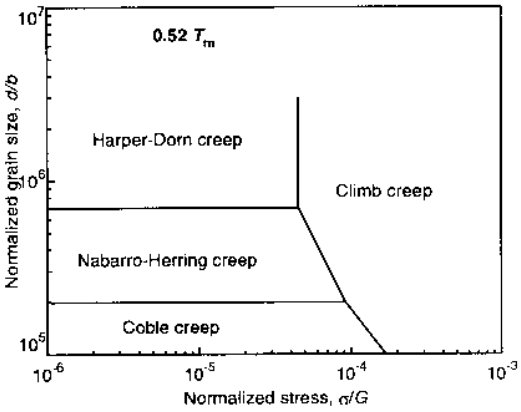
Creep Mechanisms

Pure titanium: Creep mechanism map above 0.4 T_m



A transition from grain boundary diffusion controlled (Coble) creep to lattice diffusion controlled (Nabazzo-Herring) creep occurs at about 970 K (0.5 T_m) for a grain size of ~65 μm. A similar transition is shown on the mechanism map for 69 μ grain size. Dashed line is a contour for a constant creep rate of 10^{-10}/s.
Map construction is based on an assumed room-temperature shear modulus of 49.5 GPa and a temperature coefficient of 25 MPa/K.
Source: G. Malakondaiah and P. Rama Rao, Creep of Alpha-Titanium at Low Stresses, *Acta Metall.*, Vol 29, 1981, p 1263-1275

Pure titanium: Effect of grain size on creep mechanism



d = minimum grain diameter; b = Burgers vector.
Measured creep rates in the coarse grain size and intermediate temperature regime are found to be independent of grain size and are in good agreement with the predictions of the Harper-Dorn creep empirical relationship
Source: G. Malankondaiah and P. Rama Rao, *Acta Metall.*, Vol 29, 1981, p 1263-1275

Rupture

Stress-rupture characteristics of metals have been correlated by the following relation developed by Orr *et al.* (*Trans. Amer. Soc. Metals*, Vol 46, 1954, p 113):

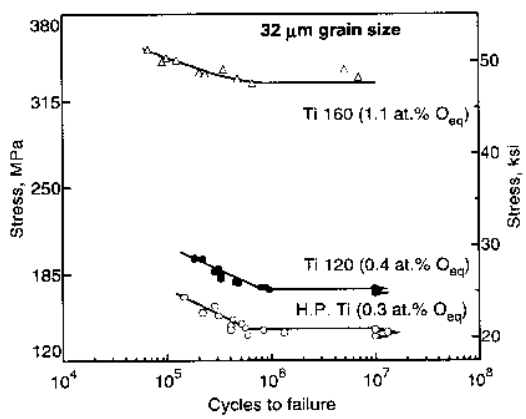
$$\sigma = f[\, t_r \exp(H/RT)\,]$$

where σ is the rupture stress, t_r is the time to rupture, and H is a constant (which is considered to be the activation energy for creep and self-diffusion in the case of pure metals). Orr *et al* assigned an activation energy of 250 kJ/mole (60 kcal/mole) for commercial purity 75A (99.6% Ti), which is considered a good estimate for high purity titanium as well.

Fatigue Properties

Fatigue limits of unalloyed titanium depend on interstitial contents, but the ratio of fatigue limits and yield strength appears relatively constant at room temperature (see figure, Conrad *et al.*, Critical Review: Deformation and Fracture, *Titanium Science and Technology*, Plenum Press, 1973, p 996-1000). The ratio of fatigue limit and yield stress does show a temperature dependence (see figure, N.G. Turner and W.T. Roberts).

Unalloyed titanium: Effect of impurities on fatigue limits



Source: N.G. Turner and W.T. Roberts, *Trans. AIME*, Vol 242, 1968, p 1223

cous creep behavior indicating grain boundary creep to be the dominant mechanism. When the applied stress exceeds a transition-stress, a switch occurs from a viscous creep mechanism to a power-law, climb-creep mechanism with a stress exponent of 4.2 to 4.6.

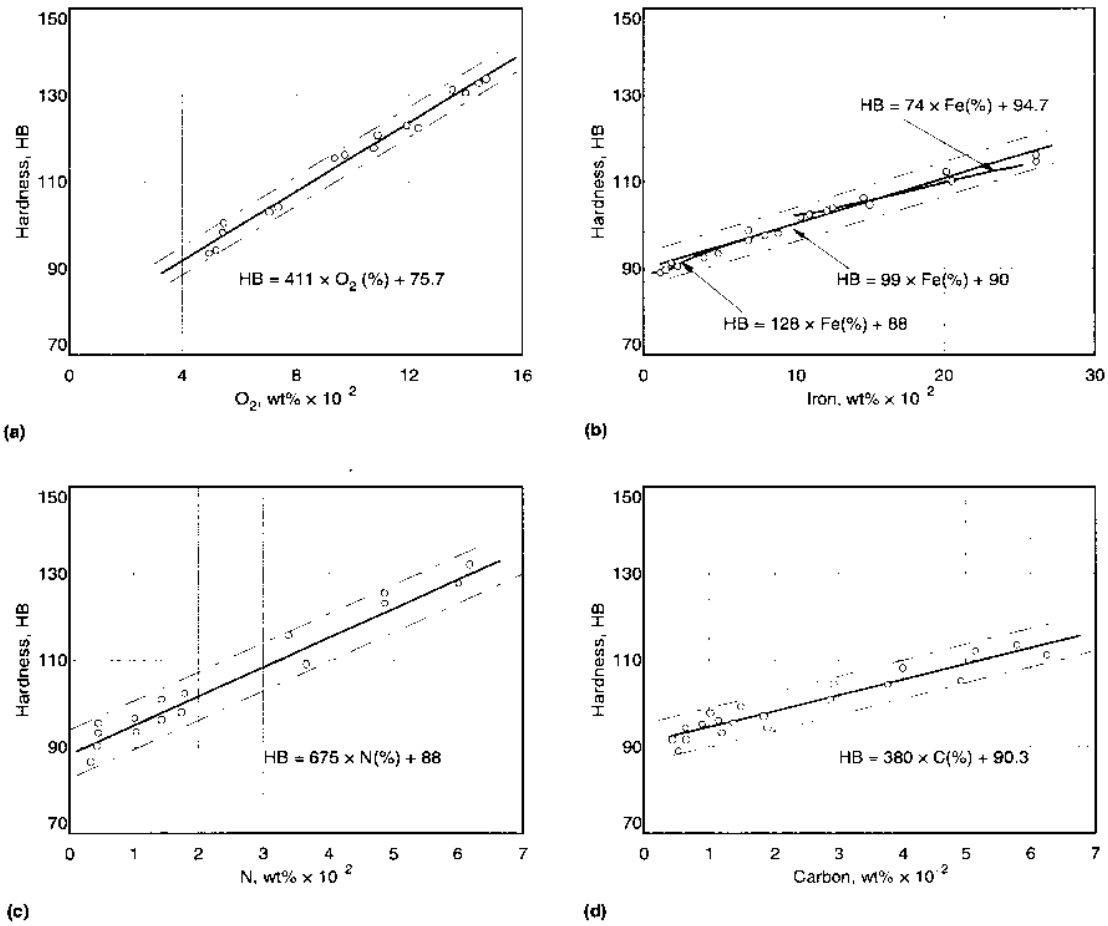
Pure titanium: Creep behavior above 0.4 T_m (500 °C)

Prior treatment of spring specimen(a)	Average grain size μm	Test temp K	Total duration of test h	Duration in transient creep h	Maximum applied stress MPa	Measured $d\varepsilon/d\sigma$ $(MPa \cdot s)^{-1}$
923 K/⅙ h	34	823	56.0	25.0	1.68	3.46×10^{-10}
1023 K/⅙ h	45	823	100.0	30.0	1.45	1.44×10^{-10}
1023 K/½ h	50	823	97.5	40.0	1.47	1.11×10^{-10}
1000 K/½ h + 25% RA + 1123 K/⅙ h	63	823	100.0	40.0	1.64	7.25×10^{-11}
1000 K/½ h + 25% RA + 1123 K/⅙ h	67	873	100.0	30.0	1.52	1.57×10^{-10}
1000 K/½ h + 25% RA + 1123 K/⅙ h	69	933	36.0	15.0	1.93	3.88×10^{-10}
1000 K/½ h + 25% RA + 1123 K/⅙ h	69	1000	23.5	10.0	1.55	1.45×10^{-9}
1000 K/½ h + 25% RA + 1123 K/⅙ h	69	1033	7.0	1.25	1.09	4.87×10^{-9}
1000 K/½ h + 25% RA + 1123 K/¼ h	75(b)	1048	8.5	...	1.51	5.54×10^{-9}
920 K/½ h + 16% RA + 1123 K/¼ h	84	1046	14.0	3.0	1.90	4.87×10^{-9}
1000 K/½ h + 25% RA + 1123 K/⅓ h	84	1042	8.25	2.5	1.37	3.79×10^{-9}
		1008	14.5	2.5	1.37	1.41×10^{-9}
		992	24.0	7.0	1.37	9.99×10^{-10}
920 K/½ h + 10% RA + 1123 K/½ h	86	1049	22.0	3.5	0.99	4.60×10^{-9}
1000 K/½ h + 15% RA + 1123 K/⅙ h	88	1000	47.7	15.0	1.27	1.16×10^{-9}
1000 K/½ h + 15% RA + 1123 K/⅙ h	91	1000	40.0	15.0	1.26	1.23×10^{-9}
920 K/½ h + 8% RA + 1123 K/½ h	110	1046	17.5	3.0	1.13	6.89×10^{-9}
1000 K/½ h + 15% RA + 1123 K/⅓ h	143	1000	30.5	10.0	1.28	1.13×10^{-9}
1000 K/½ h + 15% RA + 1123 K/½ h	195	1000	45.0	10.0	1.29	1.16×10^{-9}
1123 K/1 h	198	956	61.0	25.0	1.62	4.18×10^{-10}
1123 K/1 h	200	1058	12.0	1.0	0.98	4.41×10^{-9}
		1013	22.0	4.0	0.98	1.58×10^{-9}
1123 K/1 h	228	1058	13.0	6.0	1.39	3.21×10^{-9}
1123 K/1 h	238	1088	7.0	3.0	1.68	9.00×10^{-9}
1123 K/1 h	249	1010	23.0	9.0	1.67	1.45×10^{-9}
1100 K/1 h + 6% RA + 1143 K/2 h	276	1061	9.0	2.5	1.28	3.89×10^{-9}
		1032	11.5	1.5	1.28	1.61×10^{-9}
		1000	24.0	4.0	1.28	8.27×10^{-10}
1100 K/½ h + 15% RA + 1123 K/2 h	282	1000	24.0	7.5	1.28	1.05×10^{-9}
1100 K/½ h + 6% RA + 1123 K/1 h	315	1000	36.0	10.0	1.31	1.78×10^{-9}
1100 K/1 h + 6% RA + 1143 K/2 h	443	1000	40.0	10.0	1.25	1.67×10^{-9}

(a) To measure low strain rates ($<10^{-9}$/s) at stresses less than 2 MPa (300 psi) a sensitive spring geometry specimen was employed. RA = reduction of area by wire drawing. Initial material was 1.5 mm wire having an oxygen content of 2600 ppm. (b) During the test, grain size increased from 75 to 86 microns. Source: G. Malakondaiah and P. Rama Rao, "Creep of Titanium at Low Stresses, *Acta Metall.*, Vol 29, 1981, p 1263-1275

Ingot melted from electrolytic titanium has a typical hardness of 70 to 74 HB. Ingot from iodide titanium has hardness of 65 to 72 HB.

Pure titanium: Effect of impurities on Brinell hardness



Relation between Brinell hardness (HB) and (a) O_2 ,(b) iron ,(c) nitrogen, and (d) carbon content.
Source: K. Shimasaki *et al.*, Relation Between Brinell Hardness of Titanium and Impurities, *Titanium 80, Science and Technology*, AIME, 1980, p 1131-1136

Creep Properties

Creep Deformation

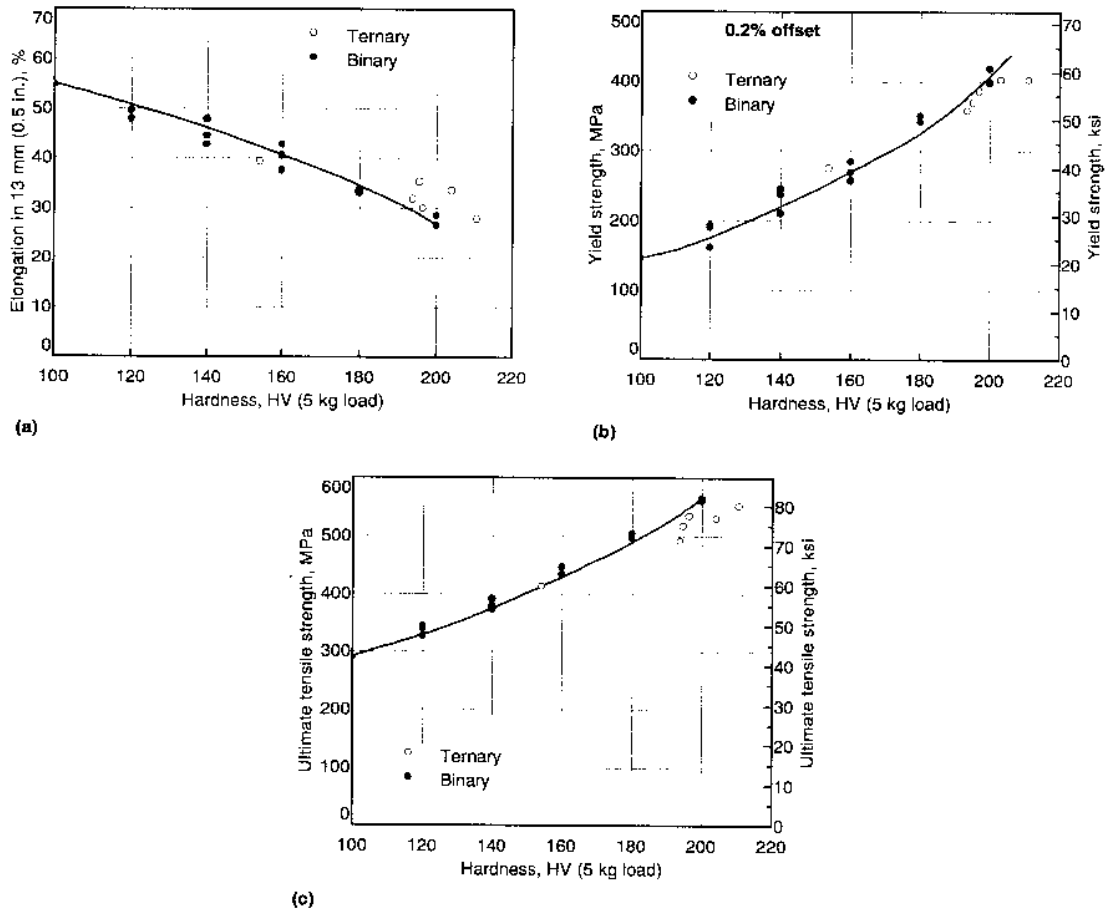
Diffusion creep of high purity (99.9%) titanium is similar to that of commercial purity titanium, and interstitial impurities have little influence on strength levels above 0.3 T_m (580 K, or 300 °C). At temperatures higher than 0.4 T_m (500 °C), titanium becomes very soft and undergoes dynamic recrystallization. Below 0.4 T_m, self-diffusion is not expected to play a role in mechanical properties.

Below 0.4 T_m. Between about 150 and 370 °C (300 and 700 °F) commercially pure titanium does not normally creep at stresses below the yield strength, although creep below 150 °C (300 °F) has been reported (*Metals Handbook*, 8th ed., Vol 1, p 537). This reversal in creep behavior may be associated with residual stress. Above 0.25 T_m (210 °C) and at stresses less than $3 \times 10^{-3} E$ (where E = Young's modulus), deformation contributes to the growth of voids, which results in transgranular creep fracture (Y. Krishnamohanrao *et al.*, *Acta Metall.*, Vol 34, No. 9, 1986, p 1783-1806).

Above 0.4 T_m. At temperatures above 0.4 T_m (500 °C), creep can occur at relative low stresses on the order of $10^{-5} G$, where G is the shear modulus. Comprehensive data of creep behavior in this regime has been generated over a wide range of grain sizes by Malakondoiah and Rama Rao (see table). Creep mechanism maps (see figures) also have been generated for ranges of stress, temperature, and grain size. Over the range of temperature (550 to 815 °C) and grain size (34 to 443 μm) investigated, titanium exhibits a Newtonian vis-

Correlation with Hardness

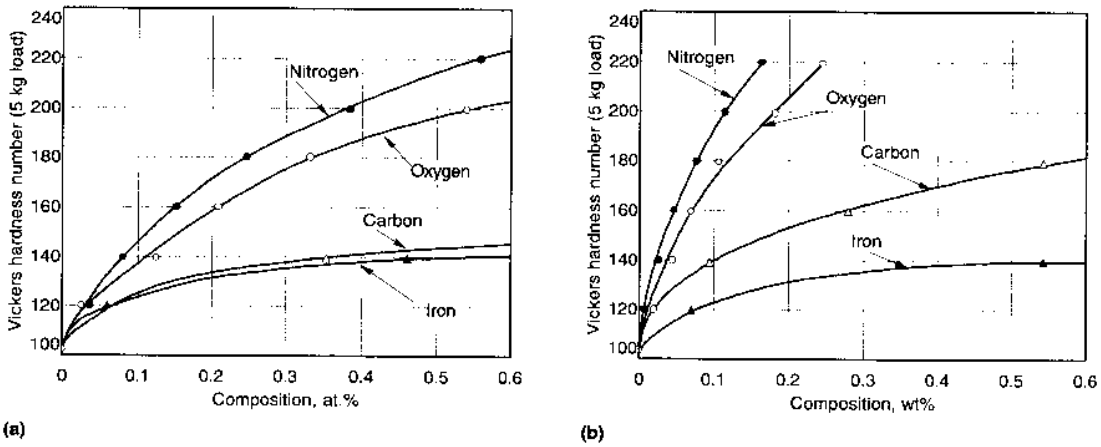
Pure Ti: Correlation of tensile properties with hardness



Impurity contents of binary alloys range from 0.04 at.% to 2.0 at.% (see the figure showing the increases in ultimate tensile strength for binary alloys). Annealed, high-purity titanium.
Source: W.L. Finlay and J.A. Synder, *J. Met.*, Vol 188, Feb 1950, p 280-282

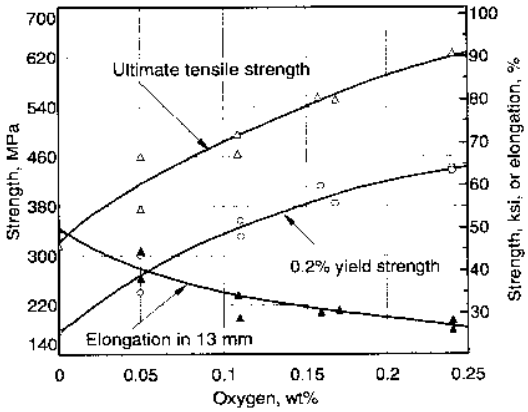
Hardness

Pure titanium: Effect of impurities on Vickers hardness



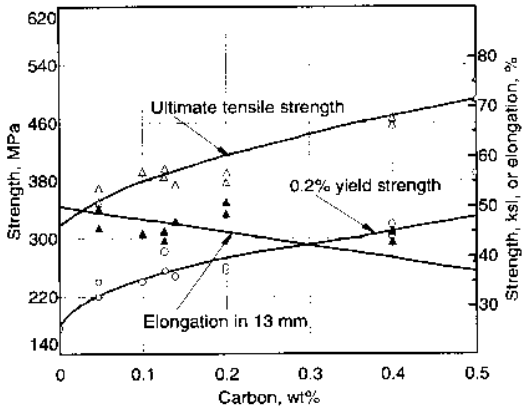
Titanium ingots having dilute carbon, nitrogen and oxygen impurity levels were cold rolled to 1 mm (0.040 in.) thick sheet, mechanically cleaned, 1000 °C vacuum annealed, cold rolled to 0.5 mm (0.020 in.), and finally were annealed for one hour at 700 °C in vacuum. This treatment gave an average equiaxed grain size of 0.020 to 0.025 mm.
Source: W. L. Finlay and J.A. Synder, *J. Met.*, Feb 1950, and *Trans. AIME*, Vol 188, p 280

Pure titanium: Effect of oxygen on tensile properties



Source: W.L. Finlay and J.A. Synder, Effects of Three Interstitial Solutes (Nitrogen, Oxygen, and Carbon) on The Mechanical Properties of High-Purity, Alpha Titanium, *Trans. AIME*, Vol 188; *J. Met.*, Feb 1950, p 277-285

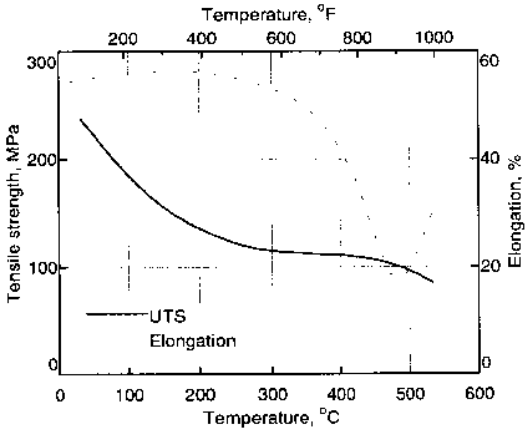
Pure titanium: Effect of carbon on tensile properties



W.L. Finlay and J.A. Synder, Effects of Three Interstitial Solutes (Nitrogen, Oxygen, and Carbon) on The Mechanical Propeties of High-Purity, Alpha Titanium, *Trans. AIME* Vol 188; J. Met., Feb 1950, p 277-285

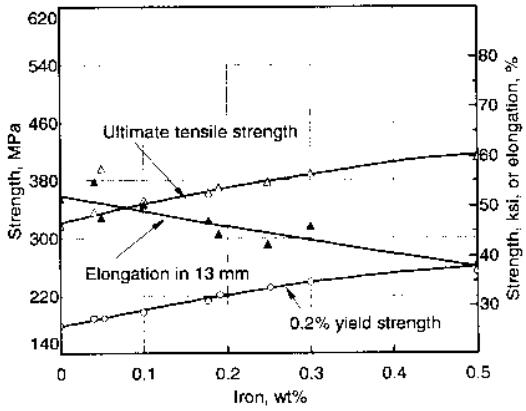
Effect of Temperature

Typical elongation and ultimate tensile strength



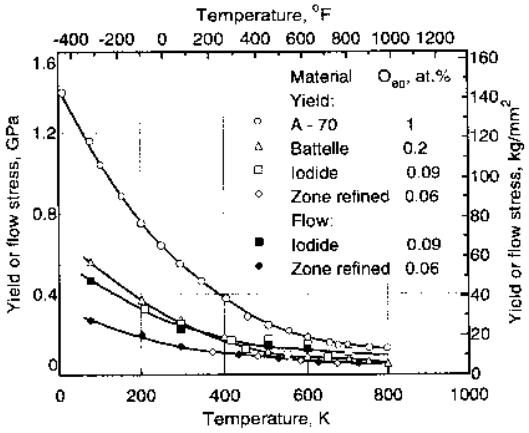
Source: *Metals Handbook*, 9th ed., Vol 2, p 815

Pure titanium: Effect of iron on tensile properties



Source: W.L. Finlay and J.A. Synder, Effects of Three Interstitial Solutes (Nitrogen, Oxygen, and Carbon) on The Mechanical Properties of High-Purity, Alpha Titanium, *Trans. AIME*, Vol 188; *J. Met.*, Feb 1950, p 277-285

Effect of temperature on tensile yield strength and flow stress



Grain sizes of 1 to 3 μm and strain rate of about 10^{-4}/s.
H. Conrad and R. Jones, *The Science, Technology and Application of Titanium*, Jaffee and Promisel, Ed., Pergamon, 1970, p 494

Tensile Properties

Typical room-temperature tensile properties of pure titanium are:

- 235 MPa (34 ksi) ultimate tensile strength
- 140 MPa (20 ksi) tensile yield strength (0.2% offset)
- 50% elongation in 50 mm (2 in.)

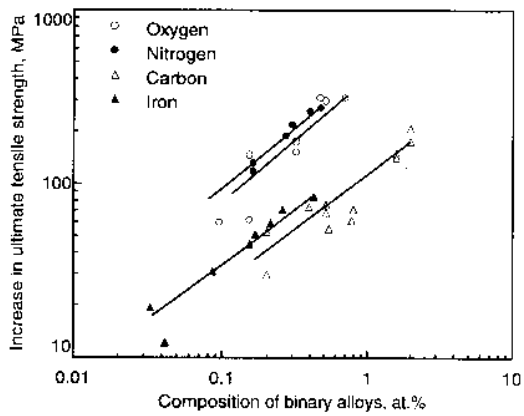
Effect of Impurities. Interstitial impurities have a strong influence on tensile properties, which can be correlated to the hardening effect of the impurities (see figures).

Grain Size Effect. Effect of grain size on tensile yield strength obeys the Hall-Petch relation, but the effect is less pronounced for the more pure grades of titanium.

IMI 110 titanium sheet: Guaranteed mechanical properties

Minimum 0.2% yield strength	130 MPa (18.8 ksi)
Ultimate tensile strength	270-350 MPa (39-50 ksi)
Minimum elongation on 50 mm (2 in.)	30%
Maximum bend radius:.	
Sheet up to 1.83 mm (0.072 in.)	1*T*
Sheet up to 1.83-3.25 mm (0.128 in.)	2*T*

Pure Ti: Effect of impurities on tensile strength



Titanium ingots having dilute carbon, nitrogen and oxygen impurity levels were cold rolled to 1 mm (0.040 in.) thick sheet, mechanically cleaned, 1000 °C vacuum annealed, cold rolled to 0.5 mm (0.020 in.), and finally were annealed for one hour at 700 °C in vacuum. This treatment gave an average equiaxed grain size of 0.020 to 0.025 mm.

Source: W.L. Finlay and J.A. Snyder, *J. Met.*, Vol 188, Feb 1950, p 280-282

Effect of Grain Size

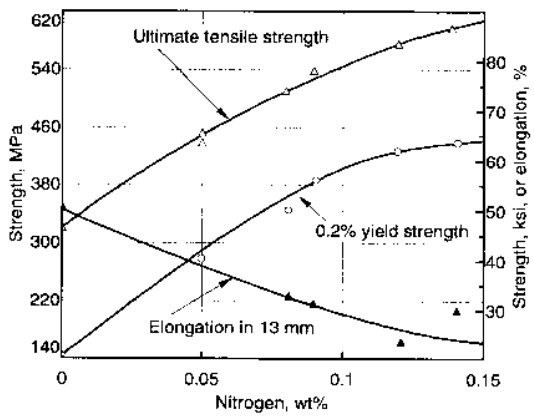
Effect of grain size on room-temperature yield strength of three purities of titanium

Material	Impurities (by weight), ppm	Hall-Petch factors(a): Friction stress (σ_i)(b) kgf/mm^2	MPa	Grain size constant (K) kgf/mm$^{3/2}$	MPa/$\sqrt{m}$
A70 (ASTM grade 4)	2830 O, 25 C, 250 N, 67 H, 3700 Fe	40	390	0.86	266
Battelle	330 O, 25 C, 140 N, 62 H, 90 Fe	18	175	0.32	99
Iodide	30 O, 18 C, 110 N, 18 H	14	137	0.47	145

(a) Factors in the Hall-Petch relation: yield stress = $\sigma_i + K\sqrt{d}$, where d is the average grain size. (b) Friction stress for a strain rate of 3×10^{-4}/s. Source: H. Conrad and R. Jones, *The Science, Technology and Application of Titanium*, Jaffee and Promise, Ed., Pergamon Press, 1970, p 493

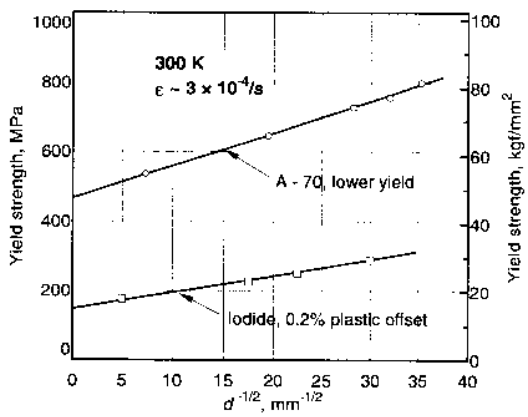
Effect of Impurities

Pure titanium: Effect of nitrogen on tensile properties



Source: W.L. Finlay and J.A. Synder, Effects of Three Interstitial Solutes (Nitrogen, Oxygen, and Carbon) on The Mechanical Properties of High-Purity, Alpha Titanium, *Trans. AIME*, Vol 188; *J. Met.*, Feb 1950, p 277-285

Effect of grain size on tensile yield strength



Room-temperature strength of ASTM grade 4 titanium (A-70) and iodide titanium (30 ppm O, 110 ppm N) tested at a strain rate of 3×10^{-4}/s.

Source: R.L. Jones and H. Conrad, *Trans. TMS/AIME*, Vol 245, 1969, p 779-789

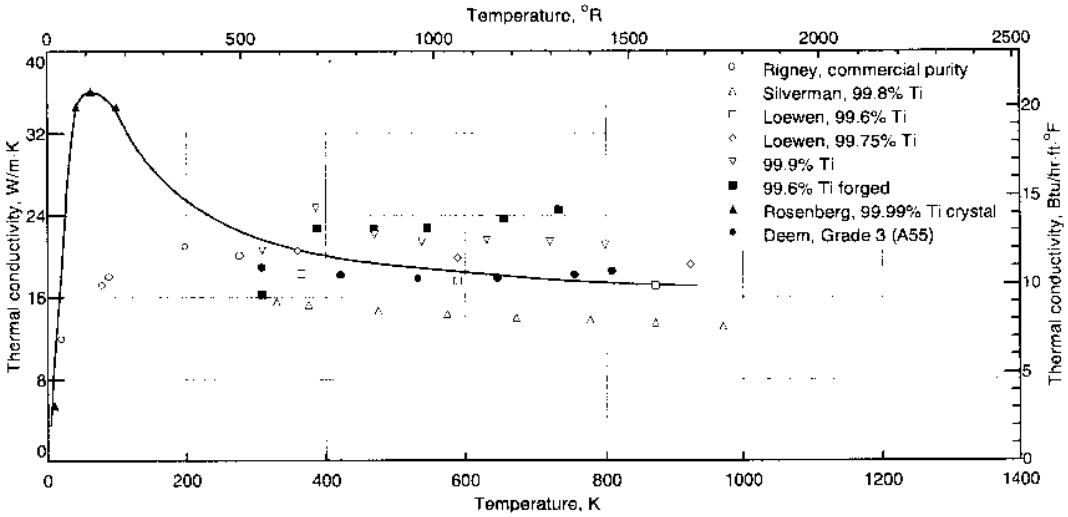
Thermal Conductivity

Thermal conductivity of various pure titanium samples

Sample description and data source footnote	Thermal conductivity, W/m·°C (Btu/ft·h·°F), at mean temperatures of: -240 °C (-400 °F)	0 °C (32 °F)	20 °C (68 °F)	25 °C (77 °F)	50 °C (120 °F)	100 °C (212 °F)	150 °C (300 °F)	200 °C (390 °F)	250 °C (480 °F)	300 °C (570 °F)
99.99% Ti(a)	19.4 (11.2)	...	...	...	...	...	...	...	...	...
99.9% Ti(a)	11.4 (6.6)	...	...	...	...	...	...	...	...	...
Polycrystalline solid with unspecified impurity(b)	...	22.4 (12.95)	...	21.9 (12.65)	...	20.7 (11.97)	...	...	...	...
Normal commercial purity(c)	...	...	...	...	18.86 (10.90)	18.75 (10.84)	18.64 (10.77)	18.6 (10.75)	18.5 (10.7)	18.4 (10.6)
High-purity grade(c)	...	...	...	...	20.5 (11.85)	20.1 (11.6)	19.7 (11.4)	19.25 (11.1)	18.9 (10.9)	18.55 (10.7)
Very high purity(c)	...	...	20.0 (12.7)	...	...	...	...	...	...	...

(a) Mendelssohn and Rosenberg, *Proceeding of Physical Society* (London), Vol 65A, 1952, p 385. (b) C.Y. Ho *et al.*, *Journal of Physical and Chemical Reference Data*, Vol 1, 1972, p 279-421. (c) E.W. Powell and R.P. Tye, *Journal of Less-Common Metals*, Vol 3, 1961, p 226-233

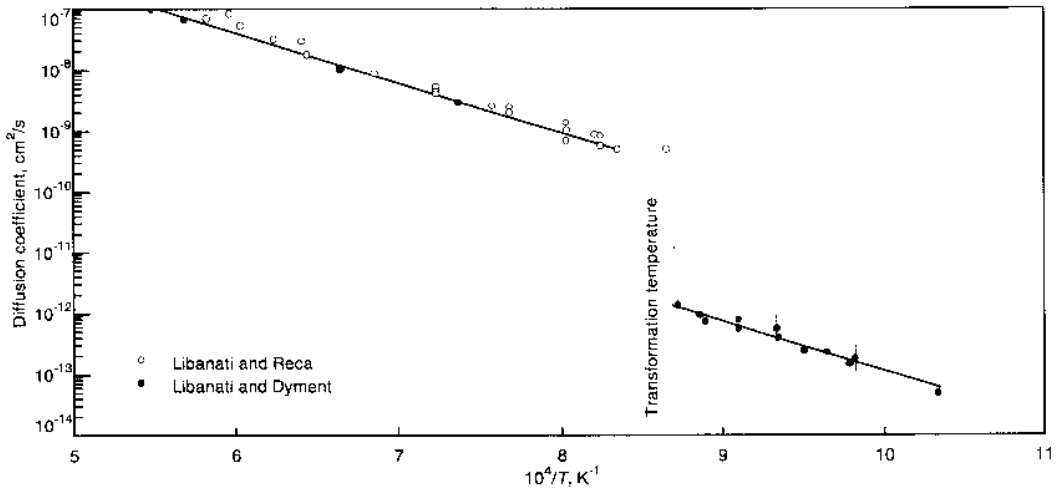
Pure Titanium: Thermal conductivity data



Source: *Thermophysical Properties of Solid Materials, Vol 1 - Elements*, WADC Technical Report 58-476, Wright Air Development Division, 1960

Mass Diffusivity

Pure titanium diffusion rates



Diffusion of Ti^{44} in alpha and beta titanium (99.99 wt%).
Source: F. Dyment and C.M. Libanati, Self-Diffusion of Ti, Zr, and Hf in Their HCP Phases, and Diffusion of Nb^{95} in HPC Zr, *J. Mater. Sci.*, Vol 3, 1968, p 349-359.

92 kJ/kg (estimated value in *Metals Handbook*, Vol 2, 9th ed.)

Latent Heat of Fusion:

295 ± 3 kJ/kg (79Jan);

288 ± 10 kJ/kg (84Des);

440 kJ/kg (estimated value in *Metals Handbook*, Vol 2, 9th ed.)

Latent Heat of Vaporization. Estimated at 9.83 MJ/kg in *Metals Handbook*, 9th ed.

Thermal Expansion

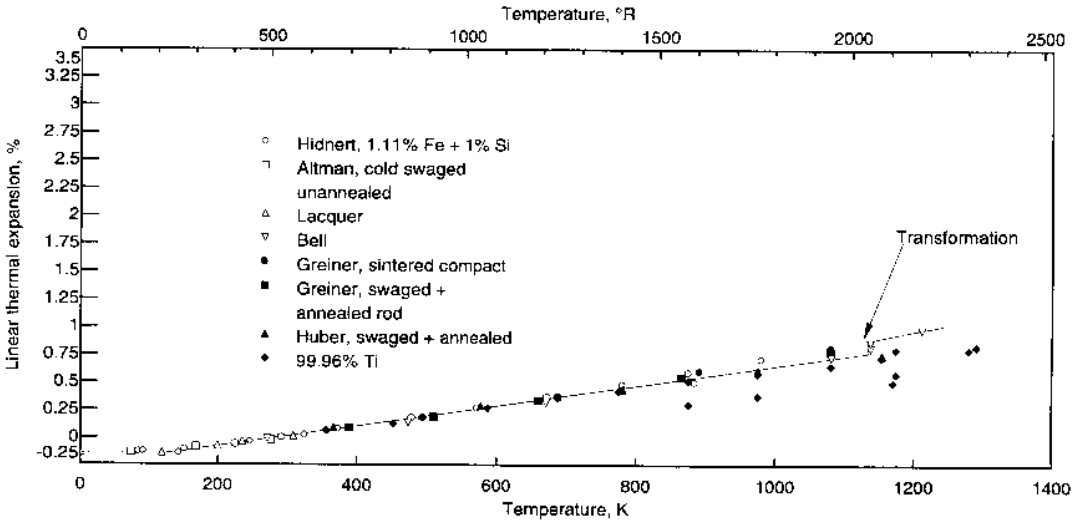
The thermal coefficient of linear expansion is about 10.8 ppm/K at 900 to 1100 °C and 14.7 ppm/K at 100 to 1500 °C (see table) based on the work of Spreadborough (59Spr) and Schmitz-Pranghe (68Sch). These expressions lead to a lattice parameter of 0.32763 nm at room temperature. This value is comparable to values obtained by extrapolation of lattice parameters of quenched binary alloys (Ti-Cr and Ti-V) to 100% Ti.

Pure titanium: Coefficients of linear thermal expansion

Reference(a)	Coefficient $10^6 \frac{da}{a} dT$,°C^{-1}	Coefficient $10^6 \frac{dc}{c} dT$,°C^{-1}	Temperature range,°C
[59Spr]	9.55 ± 0.5	10.65 ± 0.7	25 to 700
[53Mch]	11.0	8.8	25 to 225
[68Paw]	9.5	5.6	28 to 155
[53Bcr]	11.03	13.37	25 to 700
[68Sch]	~6	~8	200
	~10	~12.7	400
	~16	~18.7	882

(a) Full citations listed in *Phase Diagrams of Binary Titanium Alloys* (Murray, Ed), ASM International, 1987, p 2-4

Pure Titanium: Linear thermal expansion



Source: *Thermophysical Properties of Solid Materials, Vol 1 - Elements*, WADC Technical Report 58-747, Wright Air Development Division, 1960

Transport Properties

Titanium has a low thermal conductivity compared to that of other metals, but its heat-transfer rate is greater than that of most copper-base alloys. The reason for this has to do with favorable surface film characteristics and lack of corrosion. The major factor in heat transfer relates to material thickness, corrosion resistance, and surface films, not the thermal conductivity of the metal. In this regard, titanium has the following advantages:

- Good strength (depending on impurity levels)
- Resistance to erosion and erosion-corrosion
- Very thin, conductive oxide surface film
- Hard, smooth surface that limits adhesion of foreign materials
- Surface promotes dropwise condensation

Thermal conductivity is affected by impurity content, and data on pure titanium are limited (see table). The value for very high purity titanium (99.99 wt% Ti) at –240 °C (–400 °F), is almost double that of 99.9 wt% Ti at the same temperature.

Thermal Diffusivity. 0.022 m^2/h at 570 °C (1057 °F), 0.02 m^2/h at 700 °C (1300 °F), and 0.0185 m^2/h at 840 °C (1540 °F) per McIntosh *et al.*, *Trans ASME*, Vol 76, 1954, p 407-410.

Effect of oxygen on specific heat

K	°F	Specific heat (C_p), J/kg · °C (cal/mole · K) 99.95-99.99% pure titanium(a)	Ti-1.90 wt% O(b)	Ti-9.95 wt% O(b)
50	−370	101.1 (1.136)	91.78 (1.031)	60.8 (0.683)
75	−325	213.8 (2.402)	202.2 (2.271)	155.8 (1.750)
100	−280	(3.434)	(3.310)	(2.831)
125	−235	(4.155)	(4.074)	(3.780)
150	190	(4.684)	(4.650)	(4.564)
175	145	(5.043)	(5.067)	(5.141)
200	100	(5.321)	(5.380)	(5.697)
225	−55	(5.539)	(5.627)	(6.101)
250	10	(5.713)	(5.822)	(6.429)
275	+35	(5.864)	(5.991)	(6.703)
298.15	77	532.0 (5.976)	545.7 (6.130)	616.9 (6.930)

(a) Data from: M.H. Aven *et al.*, *Phys. Rev*, Vol 102, 1956, p 1263. (b) Data from A.D. Mah *et al.*, Thermodynamic Properties of Titanium-Oxygen Solutions and Compounds, United States Department of the Interior, Bureau of Mines, Report of Investigation 5316 (March, 1957)

Specific heat of titanium at elevated temperatures

Temperature K	°F	Specific heat (C_p) J/kg · °C	cal/mole · K
Alpha titanium ($C_p = 669.0 - 0.0371887T - 1.080 \times 10^7 T^{-2}$)(a)			
300	80	547.7	6.153
350	170	578.3	6.496
400	260	597.4	6.711
450	350	610.0	6.852
500	440	618.4	6.947
550	530	624.1	7.011
600	620	628.0	7.055
650	710	630.7	7.085
700	800	632.4	7.104
750	890	633.4	7.115
800	980	633.9	7.121(b)
850	1070	633.9	7.121
900	1160	633.7	7.119
950	1250	633.2	7.113
1000	1340	632.5	7.105
1050	1430	631.6	7.095
1100	1520	630.6	7.084
1150	1610	629.5	7.071
Beta titanium			
1150-1300	1610-1880	712	8.0

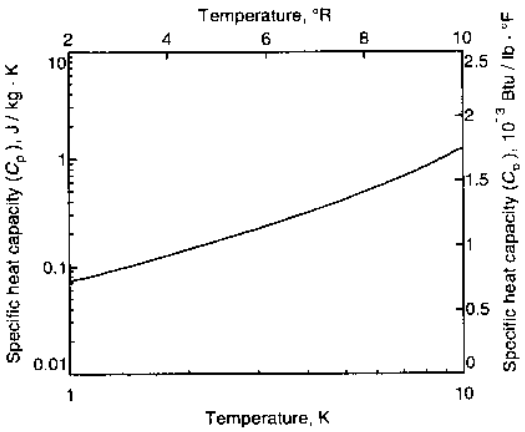
(a) Formula for specific heat (in J/kg · °C) at other temperatures (T, in K) between room temperature and the transformation temperature. (b) Anomalous maximum, which requires more work in the temperature range above 800 K (980 °F). Source: W.J. Lepkowski and J.W. Holladay, TML Report No. 73, Battelle, 1957, p 26-28

Pure Ti: Compilation of selected specific heats at low temperatures

Temperature K	°R	C_p J/kg·K	cal/lb · °F	Temperature K	°R	C_p J/kg·K	cal/lb · °F
1	1.8	0.071	0.0043	70		189	11.4
2	3.6	0.146	0.0088	80		230	13.8
3	5.4	0.226	0.0136	90		267	16.1
4	7.2	0.317	0.0191	100	180	300	18.05
6	10.8	0.54	0.0325	120		352	21.2
8	14.4	0.84	0.0505	140		391	23.5
10	18	1.26	0.0760	160		422	25.4
15		3.3	0.20	180		446	26.8
20	36	7.0	0.42	200	360	465	28.0
25		13.4	0.807	220		480	28.9
30	54	24.5	1.475	240		493	29.7
40		57.1	3.44	260		504	30.35
50		99.2	5.97	280		514	30.95
60		146.7	8.835	300	540	522	31.44

References: Data listed in WADD Technical Report 60-56 Part II (V.J. Johnson, Ed.) from the following three sources: (1) M.H. Aven *et al.*, *Phys. Rev.* Vol 102, 1956, p 1263; (2) C.W. Kothen and H.L. Johnston, *J. Am. Chem. Soc.*, Vol 75, 1953, p 1301; and (3) N.M. Wolcott, Conf. de Physique des Basses Temperatures, Paris, 1955

Pure titanium: Specific heat at 1 to 10 K



Source: V.J. Johnson, A Compendium of the Properties of Materials at Low Temperatures (Phase I), WADD Technical Report 60-56 Part II, Wright Air Development Division, 1960, p 4.141

Latent Heat

The listed values of enthalpies (latent heats) of transformation and melting are from the *Metals Handbook*, Vol 2, 9th ed., 1979, and recent assessments given by the following:

- 79Jan: D. Stull and H. Prophet, Ed., *JANAF Thermochemical Tables*, 2nd ed., NSRDS-NBS 37, 1971, Supplements in *J. Phys. Chem. Ref Data*, Vol 3, p 311-480, Vol 4, p 1-175, Vol 7, p 793-940, Vol 11, p 695-940
- 84Des: P.D. Desai, "Thermodynamic Properties of Titanium," CINDAS Report 77, 1984

Values from the Metals Handbook are only estimates.

Latent Heat of Transformation:

87 ± kJ/kg (79Jan);

80 ± 8 kJ/kg (84Des);

Thermal Properties

Melting Point

At atmospheric pressure, pure titanium has an assessed melting temperature of 1670 ± 5 °C (3038 ± 9 °F) on the 1968 International Practical Temperature Scale (IPTS). This assessed value is based on past determinations (see table) and the demonstration by Rudy and Progulski (67Rud) that their samples were not contaminated during the melting experiments by checking for agreement of the βTi ↔ αTi transformation temperatures before and after the melting point determination.

Experimental determinations of the melting point of Pure βTi

Reference(a)	Reported melting point, °C	Melting point (IPTS-68), °C
[51Han]	1720 ± 10	1723
[52Ade]	1700 ± 15	1703
[53May]	1680 ± 10	1683
[53Sch]	1660 ± 10	1663
[54Ori]	1672 ± 4	1675
[56Dca]	1660 ± 10	1663
[59Bic]	1667 ± 8	1670
[67Rud]	1668 ± 4	1671
[74Bcr]	1666 ± 4	1667
[77Cez]	1672 ± 5	1672
Assessed		1670 ± 5

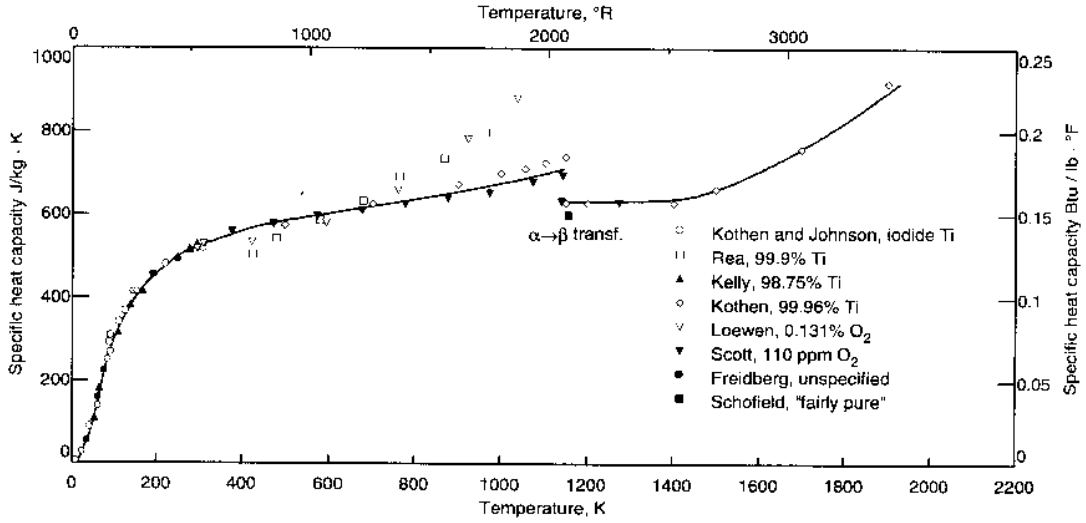
(a) Full reference citations are given in *Phase Diagrams of Binary Titanium Alloys*, J. Murray, Ed., ASM International, 1987, p 1-5

Specific Heat

Electronic Specific Heat. 0.070 J/° kg (8.0 × 10^{-4} cal/°C · mol)

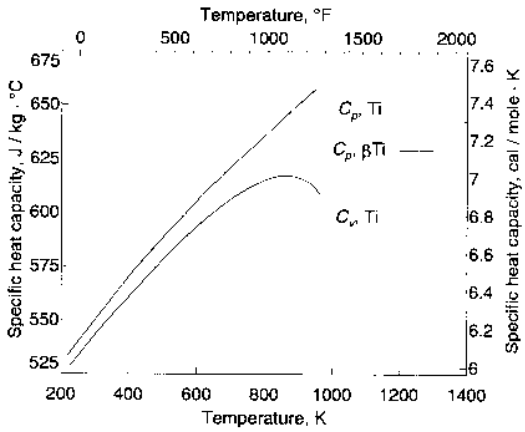
Effect of oxygen on the heat capacity (or specific heat) of pure titanium depends on the temperature range of the measurement (see table). Below about 165 K (−160 °F), titanium containing oxygen has a lower specific heat than pure titanium, but above about 165 K (−165 °F) the titanium-oxygen alloys have a higher specific heat than pure titanium. Consequently, specific heat measurements made on impure material will give values higher than the true specific heats at, or above, room temperature.

Pure titanium: Specific heat vs temperature



Source: *Thermophysical Properties of Solid Materials, Vol 1 - Elements*, WADC Technical Report 58-476, Wright Air Development Division, 1960

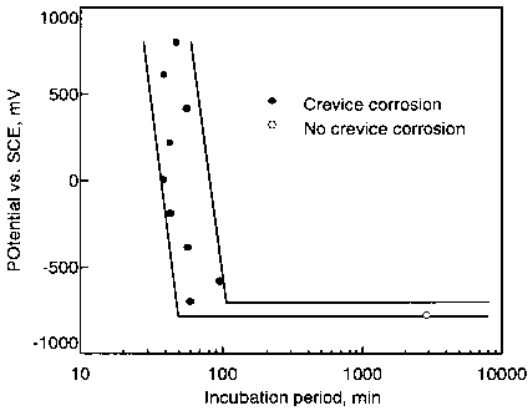
Heat capacity of pure titanium



Source: E.S. Fisher and J. Renken, *Phys. Rev.*, Vol 135, No. 2A, 1964, pA482-A494

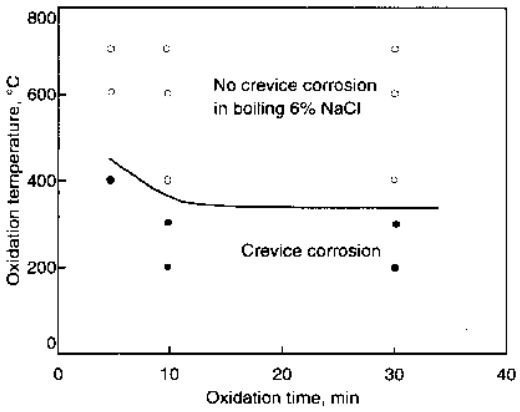
Crevice Corrosion

Crevice corrosion in boiling 6% NaCl solution



Specimens were machined to 15 or 30 mm (0.6 or 1.2 in.) squares with a 5.5 mm (0.215 in.) diam hole. Composition (wt%): 0.05 O, 0.004 C, 0.03 Fe, 0.014 N, 0.0003 H.
Source: M. Kobayashi *et al.*, *Titanium '80 Science and Technology*, The Metallurgical Society of AIME, 1980, p 2613-2622

Effect of oxidation on crevice corrosion in NaCl solution



Specimens were machined to 15 or 30 mm (0.6 or 1.2 in.) squares with a 5.5 mm (0.215 in.) diam hole. Composition (wt%): 0.05 O, 0.004 C, 0.03 Fe, 0.014 N, 0.0003 H.
Source: M. Kobayashi *et al.*, Study on Crevice Corrosion of Titanium, *Titanium '80 Science and Technology*, The Metallurgical Society of AIME, 1980, p 2613-2622

Hydrogen Damage

Titanium and its alloys suffer hydrogen damage primarily by hydride-phase formation. Uniaxial tensile properties of high-purity titanium are relatively unaffected by typical concentrations (<200 ppm) of hydrogen; however, the impact toughness can be impaired.

Commercially pure titanium is more sensitive to hydrogen than pure titanium is. The amount of hydrogen necessary to induce ductile-to-brittle transition behavior in commercially pure titanium is one-half the amount needed in pure titanium.

At room temperature, the solubility limit of hydrogen in α-titanium ranges from 20 to 200 ppm (wt%) depending on the impurity content. Hydrogen solubility in high-purity titanium is about 0.1 at.% (0.02 wt%) at room temperature (see the section "Phase Diagrams" in this datasheet).

Tensile properties, hardnesses, and impact strengths of three titanium alloys at two hydrogen levels

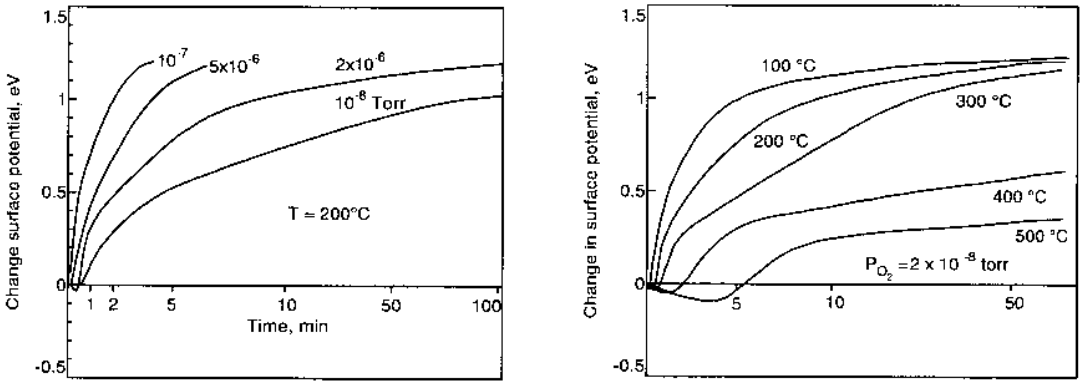
Alloy Addition, wt%	Hydrogen Atomic %	Hydrogen wt%	Treatment	Tensile strength, ksi	0.2% Offset yield strength, ksi	Elongation, %	Reduction in area, %	Room-temperature impact strength, in.-lb
0.06 N	0.07	0.0014	Vacuum annealed 775 °C 2 hr, furnace cooled	82	72	36	55	24
0.06 N	0.63	0.0133	700 °C 1 hr in argon, furnace cooled	77	63	30	57	4
0.18 N	0.09	0.0018	Vacuum annealed 775 °C 2 hr, furnace cooled	114	110	23	26	16
0.18 N	0.6	0.0127	700 °C 1 hr in argon, furnace cooled	122	117	25	23	3
2.55 Al	0.07	0.0015	Vacuum annealed 775 °C 2 hr, furnace cooled	93	78	26	47	41
2.55 Al	0.6	0.0129	900 °C 1 hr in argon, furnace cooled	96	82	23	39	7

Source: G.A. Lenning, C.M. Craighead, and R.I. Jaffee, Constitution and Mechanical Properties of Titanium-Hydrogen Alloys, *Hydrogen Damage*, C.D. Beachem, Ed., American Society for Metals, 1977, p 100

Stress-Corrosion Cracking

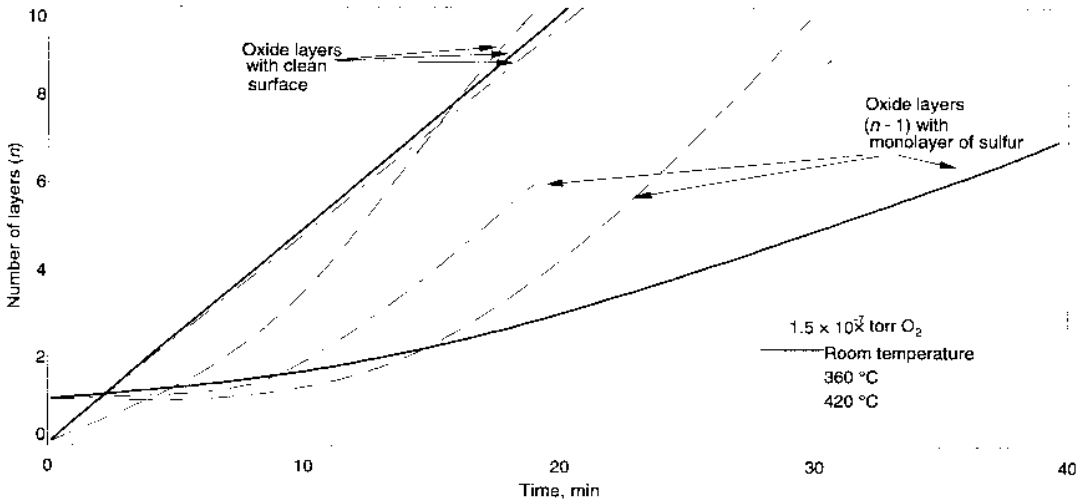
Pure titanium (and commercially pure ASTM grades 1, 2, 7, 11, and 12) is immune to stress-corrosion cracking (SCC), except in a few environments such as anhydrous methanol/halide solutions, nitrogen tetroxide (N_2O_4), red fuming HNO_3, and liquid or solid cadmium. The general SCC behavior of titanium and titanium alloys is described in "Technical Note 2: Corrosion" in this Volume and in the datasheet "CP and Modified Titanium."

Surface potential vs time at various pressures



The adsorption of oxygen is related to surface potential measurements. Titanium sample had 40 ppm O and was surface cleaned by ion bombardment in a vacuum chamber.
Source: J.B. Bignolas *et al.*, A Study of Oxygen Chemisorption on Titanium by Auger Analysis and Mirror Electron Microscopy, *Titanium '80 Science and Technology*, Metallurgical Society of AIME, 1980, 2829-2837

Oxidation kinetics of high-purity titanium



Samples were iodide titanium cleaned by successive treatments of argon ion bombardment (500 eV).
Source: J.P. Gspann *et al.*, Oxidation Kinetics of High-Purity Titanium, Influence of a Monolayer of Chemisorped Sulfur, *Titanium '80 Science and Technology*, The Metallurgical Society of AIME, 1980, p 2839-2851

deteriorate and dissolve, and the unprotected metal is oxidized to the soluble trivalent ion ($Ti^{3+} + 3e^-$). Corrosion for commercially pure titanium is covered in the article "CP and Modified Ti".

Electrochemical Potentials

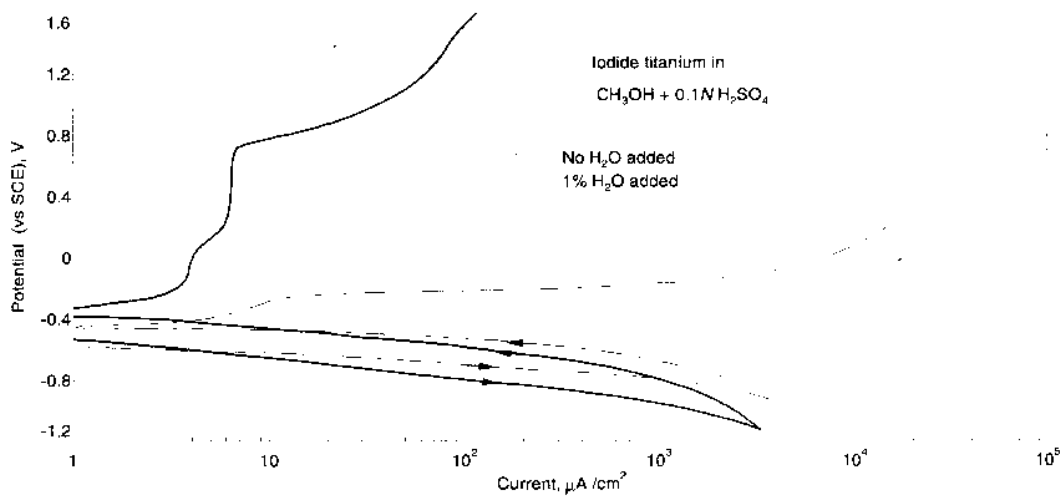
Electrochemical anodic and cathodic polarization testing is often used to supplement weight loss testing. Polarization testing can identify whether the alloy is truly fully passive or possibly metastable; this is often not discernible from weight loss tests alone.

Repassivation potentials generally decrease as iron content increases in titanium. For iron levels less than 0.3%, this reduction is considered to be insignificant, on the order of several tenths of a volt. In hot HCl solutions, for example, the useful passive region is reduced when iron levels exceed 0.3 wt% in base metal and 0.2 wt% in weldments. (Schutz *et al.*, *Titanium Science and Technology*, 1985, p 2617-2624).

The steady-state electrode potential in flowing seawater is about 0.10 V vs SCE at 24 °C (75 °F).

Polarization curves in a solution of methyl alcohol and sulfuric acid (CH3OH + H2SO4) were obtained for pure (iodide) titanium (see figure) and no significant differences were observed for Ti-75A (ASTM Grade 4) and Ti-6Al-4V. Without water additions, a short region of passivity can be detected, but a sharp increase of currents occurs at about −0.24V. Examination revealed attack in the form of pitting (see figure reference for details).

Pure titanium: Effect of water additions on potentiostatic polarization



When 1% of water was added to the test solution, the titanium electrode apparently was in the passive state. The anodic current remained small and constant up to about +0.7 V. At higher potentials the current increased steadily, but not as sharply as in water-free solutions.
Source: F. Mansfield, Pitting Caused by Chlorides or Sulfates in Organic Media, *Galvanic and Pitting Corrosion-Field and Laboratory Studies*, ASTM STP 576, ASTM, 1976, p 180-202

Oxidation

Oxidation of pure titanium does not differ significantly from that of ASTM commercially pure titanium grades 1 and 2 (UNS R50250 and R50400), and thin oxide films form in air temperatures between 315 and 650 °C (600 and 1200 °F). The oxide film is barely perceptible after exposure at 315 °C (600 °F), but it becomes darker and thicker with increasing temperature and time at temperature. The changes in surface color can be used as a rough guide of exposure temperature in air (see table), while time at temperature becomes a more significant factor for temperatures above 500 °C (see figure). Care must be used in estimating temperature due to the time factor mentioned, and the oxide color at a given temperature is strongly influenced by the cleanliness of the surface. A surface exposed to elevated temperature with, for example, oil on the surface, will have a different color than a clean surface.

The oxidation of pure titanium has also been studied at low oxygen pressures, so the first steps of oxidation can be evaluated by surface potential measurement, Auger analysis, or x-ray photoelectron microscopy (ESCA). The influence of oxidation on surface potentials is illustrated by typical pressure-temperature effects (see figure), while the growth of oxide layers have been evaluated by Auger analysis (see figure showing the effect of a sulfur monolayer on oxidation layers).

Effect of air temperature on titanium's appearance

Air temperature °C	°F	Appearance of oxide film(a)
370	700	Straw yellow color
480	900	Blue color
650	1200	Dull gray

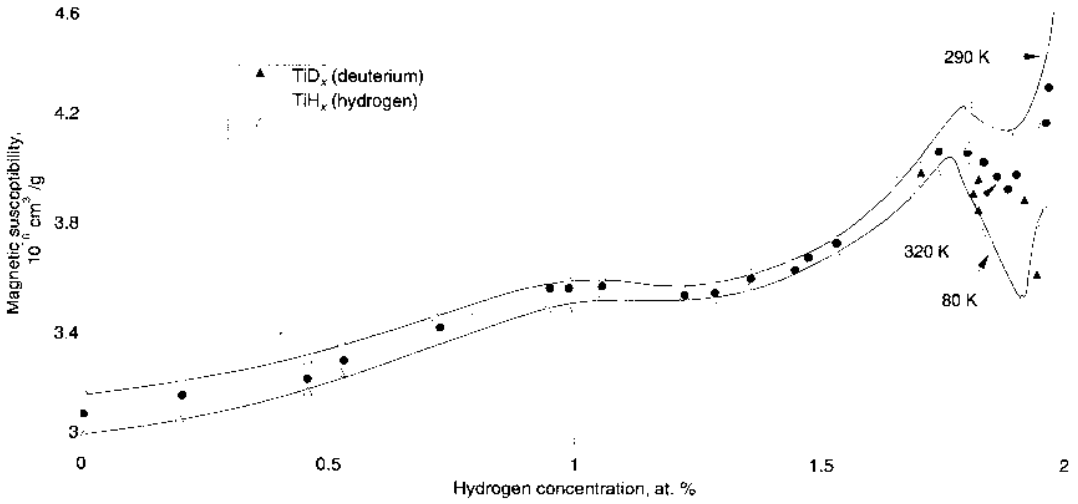
(a) Alloying elements and surface contaminants also influence color.

Magnetic Effects

Magnetoresistance Coefficient. 1.04×10^{-16} m^2/A^2 (6.6×10^{-13} Oe^{-})

Hall Coefficient. Nonmetallic impurities greatly affect the Hall coefficient. Positive Hall coefficients for pure titanium have ranged from +1 $\times 10^{-4}$ nΩ cm/Oe to a high value of +3 $\times 10^{-4}$ n^2 cm/Oe reported by G.W. Scovil ("The Hall Effect in Titanium," *J. App. Phys.*, Vol 24, 1953, p 226). Negative Hall coefficients have been measured in specimens containing 0.34 wt% oxygen (S. Foner, *Phy. Rev.*, Series 2, Vol 91, 1953, p 447).

Magnetic susceptibility of titanium-hydrogen system



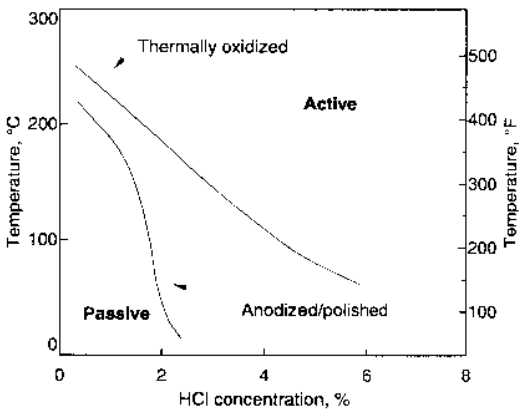
Source: W.J. Lepkowski and J.W. Holladay, The Physical Properties of Titanium and Titanium Alloys, TML Report No. 73, Battelle, 1957

Chemical/Corrosion Properties

The corrosion resistance of titanium depends on the formation of a protective oxide layer as described in "Technical Note 2: Corrosion" in this volume. The effectiveness of this layer seems to require the presence of either water or hydroxyl (even in trace amounts) during its formation. If titanium is exposed to a vigorously oxidizing environment in the complete absence of moisture, any surface film formed will not be protective and oxidation in depth may take place, often in the form of a violent exothermic reaction. Breakdown of the passive layer can also occur from dry oxidants, nonoxidizing aqueous environments as defined by a Pourbaix diagram, and pitting or crevice attack in near-neutral aqueous solutions (particularly in the presence of halides). Iron impurities, generally thought to detrimentally affect titanium corrosion, have little or no effect of practical significance at levels up to 0.2 wt% Fe (R.W. Schutz *et al.*, Effect of Solid Solution Iron on the Corrosion Behavior of Titanium, *Titanium Science and Technology*, Deutsche Gesellschaft für Metallkunde e.V., 1985, p 2617-2624.)

General corrosion becomes a concern in reducing acid environments, particularly as acid concentration and temperature increase (see figure). In strong and/or hot reducing acids (in the absence of inhibitors), the oxide film of titanium can

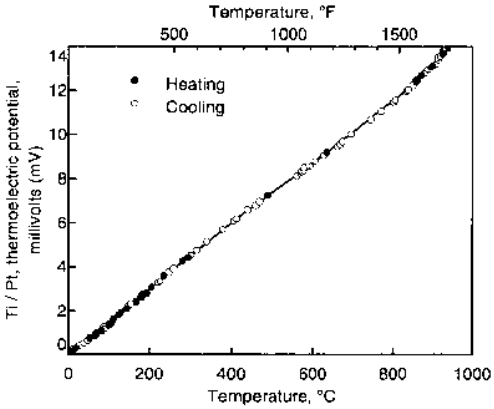
Variation in active and passive regions in HCl solution



Three types of titanium specimens were used; specimens polished with #400 emery paper, specimens anodized in a 2% phosphoric acid solution at 50V for 10 min. after polishing, and specimens oxidized in air at 600 °C for 10 min. after polishing. The thicknesses of all of these specimens were 1 mm.
Source: T. Fukuzuka *et al.*, On the Beneficial Effect of the Titanium Oxide Film Formed by Thermal Oxidation, *Titanium '80 Science and Technology*, The Metallurgical Society of AIME, 1980, p 2781-2792

Thermoelectric Properties

Thermoelectric potentials of titanium-platinum couple



Source: H.W. Womer, Thermoelectric Properties of Titanium with Special Reference to the Allotropic Transformation, *Austral, J. Sci. Res.*, Vol 4 (No. 62), 1951

Magnetic Characteristics

Magnetic Properties

Magnetic Susceptibility. 3.17×10^{-6} ($\pm 0.03 \times 10^{-6}$) cm^3/g is the mass susceptibility (χ_m) established by Lepkowski and Holladay (TML Report No. 73, Battelle, 1957). This is in good agreement with more recent measurements (see tables). Additional information on the factors affecting magnetic susceptibility are discussed in the chapter "Physical Properties" of Section 1 in this Volume.

Magnetic Permeability. 1.00005 at 1600 A/m (20 Oe)

Magnetic susceptibility of polycrystalline titanium

Specimen purity, wt% Ti	Temperature, K	Mass of specimen, mg	Estimated Number of grains per specimen	Condition	Magnetic susceptibility, 10^{-6} cm^3/g χ_1	χ_2	χ_1-χ_2	χ_{av}(a)
99.8	300	81	800	As cast	3.16_8			
99.8	300	84	800	As cast	3.16_8			
99.9	300	84	...	As received	3.16_9			
99.8	300	78	800	As cast	3.15_9	3.15_6	3.15_3	3.15_6
99.8	300	73	800	As cast	3.19_8	3.12_9	3.12_7	3.15_1
99.8	300	108	43,000	Recrystallized	3.19_3	3.14_9	3.13_3	$\underline{3.15}_8$
99.9	299	70	...	As received	3.20_0	3.12_1	3.11_0	3.14_7
99.8	299	51	53,000	Recrystallized	3.25_6	3.13_5	3.10_0	3.16_3
99.8	298	49	230,000	Recrystallized	3.35_3	3.08_5	3.07_8	$\underline{3.17}_2$

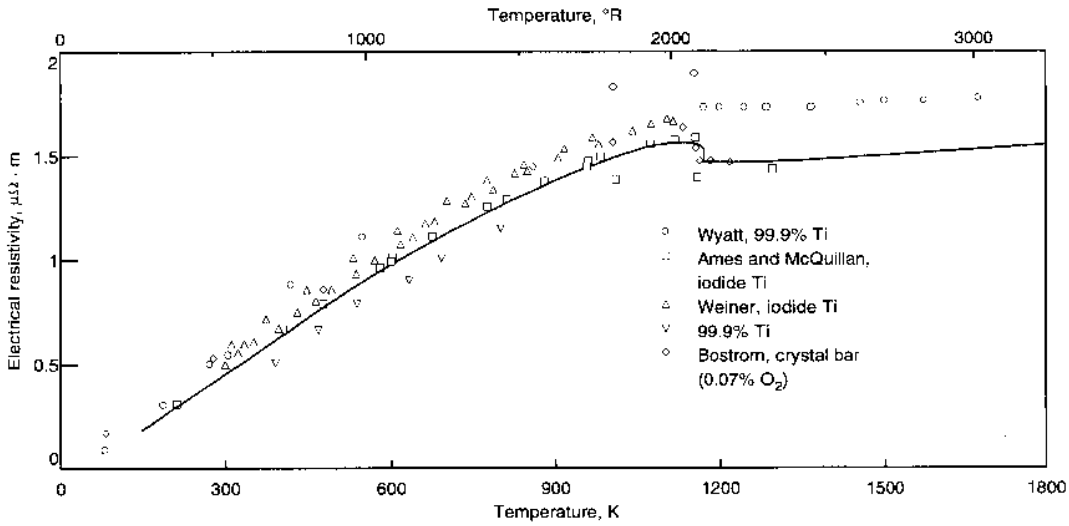
(a) Accuracy is limited to about three significant digits (as indicated by underlining). Source: E.W. Collings, Battelle, Columbus, Ohio

Magnetic susceptibility of single-crystal titanium

Magnetic susceptibility, 10^{-6} cm^3/g χ_1	χ_2	χ_1–χ_2	χ_{av}(a)
3.50_5	3.02_3	0.48_2	$\underline{3.18}_4$
3.47_3	3.02_0	0.45_3	$\underline{3.17}_1$
3.49_9	2.98_8	0.51_1	$\underline{3.15}_8$
3.52_7	3.00_6	0.52_1	$\underline{3.18}_0$

(a) Accuracy is limited to about three significant digits (as indicated by underlining). Source: E.W. Collings, Battelle, Columbus, Ohio

Pure titanium resistivity data



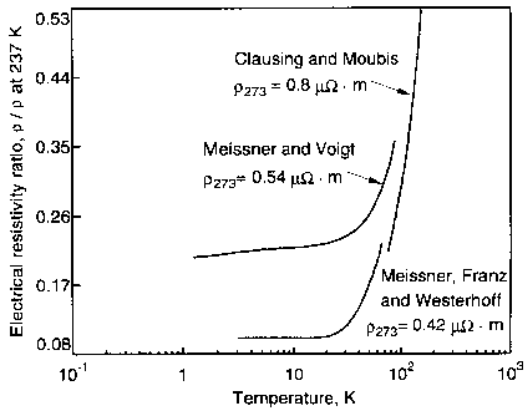
Source: *Thermophysical Properties of Solid Materials, Vol 1 - Elements*, WADA Technical Report 58-746, Wright Air Development Division, 1960

Relative resistance of a single crystal and a polycrystalline titanium sample

Temperature, K	Relative resistivity(a) Single crystal Parallel to c-axis	Perpendicular to c-axis	Polycrystalline
4.2	...	...	0.0306
19.6	...	...	0.0309
77	0.1786	0.2086	0.15
196	0.6355	0.6450	0.67
273	1.000	1.000	1.000(b)
300	1.130	1.110	1.120
373	1.4997	1.4550	1.468
400	1.64	1.58	1.60
500	2.13	2.10	2.09
600	2.60	2.60	2.56
700	2.99	2.94	2.99
800	3.25	3.17	3.31
900	3.46	3.38	3.58
1000	3.59	3.58	3.80
1100	3.67	3.76	3.98
1200	3.43	3.61	3.705
1300	3.47	3.675	3.760
1400	3.525	3.75	3.835
1450	3.555	3.78	3.865

(a) Relative resistivity = resistivity at indicated temperature/resistivity at 273 K. (b) Electrical resistivity at 273 K (0 °C) is 42.67 ± 0.05 microhm-cm. Source: James K. Stanley, *Electrical and Magnetic Properties of Metals*, American Society for Metals, 1963, p 51

Pure titanium: Resistivity at cryogenic temperatures



ρ_{273} = resistivity at 0 °C (273 K).
Source: R.B. Stewart and V.J. Johnson, Ed., *A Compendium of the Properties of Materials at Low Temperatures (Phase II)*, WADD Technical Report 60-56 (Part IV), U.S. Air Force, 1961

Superconductivity

A state of superconductivity is stable only if temperature, magnetic field strength, and current density are all below the critical levels, or thresholds. The three critical parameters of temperature (T_c), magnetic field (H_c), and current density (J_c), are closely interdependent. For example, the Hc decreases with increasing temperature or current.

Critical Temperature of Superconductivity. A typical value for pure titanium is 0.40 ± 0.04 K (B.W. Roberts, Properties of Selected Superconducting Materials, 1978 Supplement, NBS Technical Note 983). Critical temperatures as high as 0.56 K have been reported (T.S. Smith and J.G. Daunt, Phy. Rev., Series 2, Vol 88, 1952, p 1172).

Critical Field Strength of Superconductivity. Pure metals are in the class of type I superconductors, which exhibit perfect diamagnetism up to some critical field (Hc) when temperature is below Tc. Values have ranged from 2×10^{-3} T (T.S. Smith *et al.*, *Phy. Rev.* Series 2, Vol 89, 1953, p 654) to 8.6×10^{-3} T for unannealed, cold-swaged wire (M.C. Steel and R.A. Hein, Phy. Rev., Series 2, Vol 92, 1953, p 243).

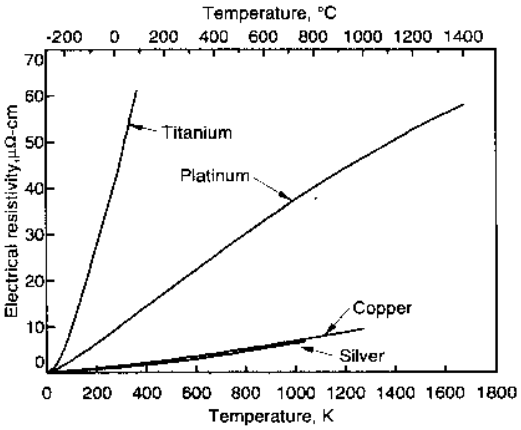
Critical current density of superconductivity depends on the degree of deformation and cooling rates during thermomechanical processing. References include E.W. Collings, Sourcebook of *Titanium Superconductivity, and Applied Superconductivity, Metallurgy, and Physics of Titanium Alloys* (Vol 1 and 2).

Electrical Properties

Resistivity

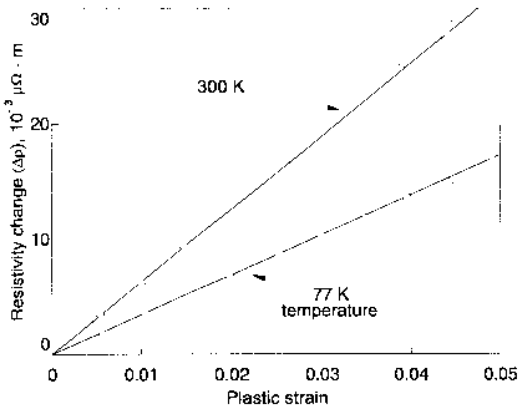
Resistivity values of 0.46 to 0.48 μΩ · m are usually obtained for iodide titanium at room temperature, although values as low as 0.42 μΩ · m have been obtained. The true resistivity of titanium has been assessed at 0.42 μΩ · m by W.J. Lepkowski and J.W. Holladay (TML Report No. 73, Battelle, 1957). Impurities increase resistivity (see the section "Physical Metallurgy of Titanium Alloys" in this volume).

Effect of temperature on the resistivity of copper, silver, platinum and titanium



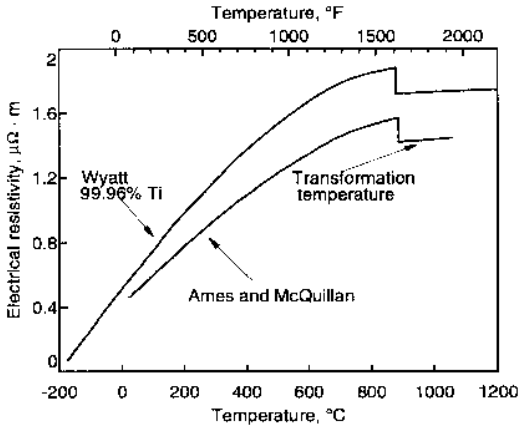
Source: J.K. Stanley, *Electrical and Magnetic Properties of Metals*, American Society for Metals, 1963

Change in resistivity with deformation



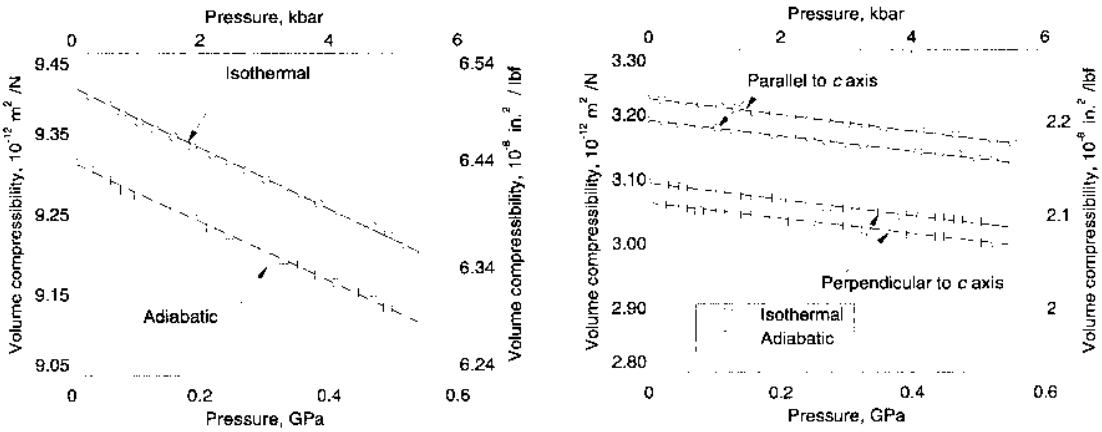
Source: F.F. Laurent 'Yev, and V.N. Nikiforenko, "Influence of Plastic Deformation and Temperature on the Resistivity of Polycrystalline Titanium," *Fiz. Met. Metalloved.*, 52 (No. 6), 1981, p 1200-1204

Effect of temperature on resistivity on high-purity titanium



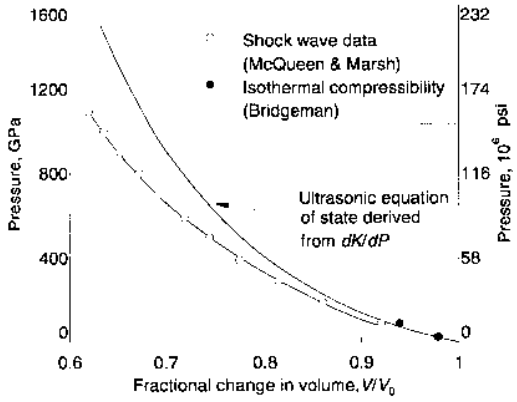
Source: J.L. Wyatt, Electrical Resistance of Titanium Metal, *J. Met.*, Vol 5, 1953, p 903-905; S.L. Ames and A.D. McQuillan, The Resistivity-Temperature Concentration Relationships in the System Niobium-Titanium, *Acta Metall.*, Vol 2, 1954, p 831

Linear and volume compressibility of single crystal titanium vs pressure



(a) Volume compressibility. (b) Linear compressibility. Material was a single crystal (zone refined from crystal iodide bar) titanium having an adiabatic bulk modulus of 107.3 GPa (15.56×10^6 psi) and an isothermal bulk modulus of 106.3 GPa (15.42×10^6 psi) at 25 °C (77 °F) and 1 bar (~1 atmosphere)
Source: E.S. Fisher and M.H. Manghani, Effects of Changes in Volume and c/a Ratio on the Pressure Derivatives of the Elastic Moduli of HCP Ti and Zr, *J. Phys. Chem. Solids*, Vol 32, 1971, p 657-667

Comparison of compression data for titanium



There is a fairly good agreement between the isothermal compressibility data of Bridgman (*Proc. Am. Acad. Arts Sci.*, Vol 76, 1958, p 55) and the ultrasonic equation. There is a poor agreement between the latter and the shockwave data (*Phys. Rev.*, Vol 108, 1957, p 196) probably because of the phase change near 9 GPa (1.3×10^6 psi). dK/dP = pressure derivative of the bulk modulus (K).
Source: E.S. Fisher and M.H. Manghani, Effects of Changes in Volume and c/a Ratio on the Pressure Derivatives of the Elastic Moduli of HCP Ti and Zr, *J. Phys., Chem. Solids*, Vol 32, 1971, p 657-667

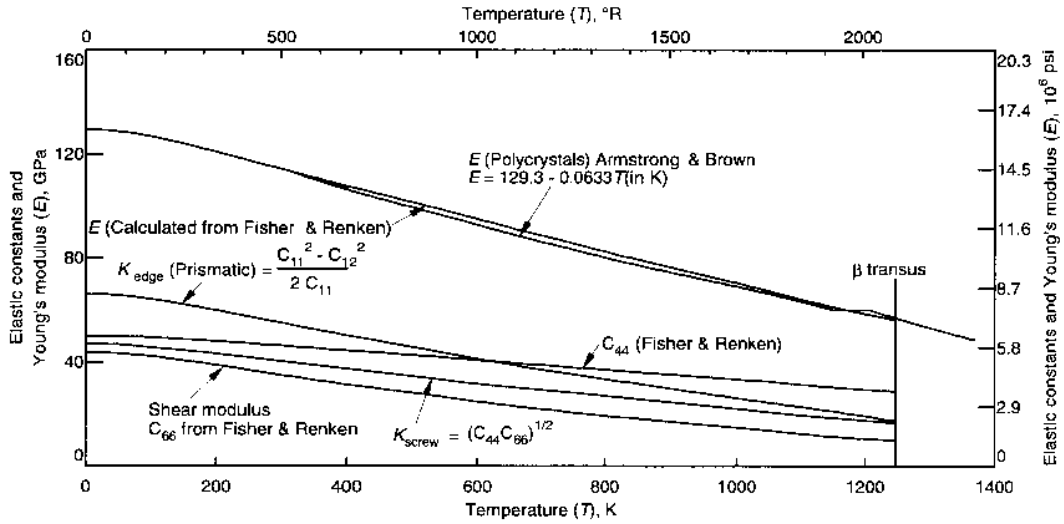
psi) for iodide titanium at room temperature and atmospheric pressure would correspond to a compressibility of about 1×10^{-11} m^2/N (7.7×10^{-8} in.^{2}/lbf). A low compressibility factor of 8.2×10^{-12} m^2/N (0.8×10^{-6} cm^2/kgf), has been reported from early work of Bridgman (*Proc. Am. Acad. Sci.*, Vol 64, 1929, p 51), while higher compressibilities are reported for iodide titanium (see figures).

Compressibility (κ) of single-crystal titanium

Temp., K	κ Perpendicular to c 10^{-12} m^2/N	κ Perpendicular to c 10^{-12} cm^2/dyne	κ Parallel to c 10^{-12} m^2/N	κ Parallel to c 10^{-12} cm^2/dyne	κ vol 10^{-12} m^2/N	κ vol 10^{-12} cm^2/dyne
4	0.300	3.00	0.309	3.09	0.908	9.08
73	0.300	3.00	0.310	3.10	0.911	9.11
173	0.303	3.03	0.314	3.14	0.920	9.21
298	0.306	3.06	0.318	3.18	0.931	9.31
373	0.309	3.09	0.322	3.22	0.940	9.40
473	0.313	3.13	0.326	3.26	0.951	9.51
573	0.316	3.16	0.330	3.30	0.963	9.63
673	0.321	3.21	0.334	3.34	0.975	9.75
773	0.326	3.26	0.337	3.37	0.988	9.88
873	0.331	3.31	0.340	3.40	1.001	10.01
973	0.336(a)	3.36(a)	0.343(a)	3.43(a)	1.016(a)	10.16(a)
1073	0.342(a)	3.42(a)	0.345(a)	3.45(a)	1.029(a)	10.29(a)
1156	0.347(a)	3.47(a)	0.346(a)	3.46(a)	1.041(a)	10.41(a)

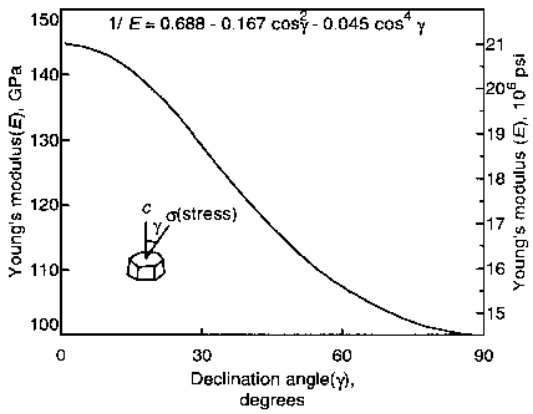
(a) Extrapolated from curves. Source: E.S. Fischer and C.J. Renken, *Physical Review*, Vol 135, No. 2A, 1964, p A482-A494

Commercially pure (>99.8%) titanium: Effect of temperature on various elastic constants



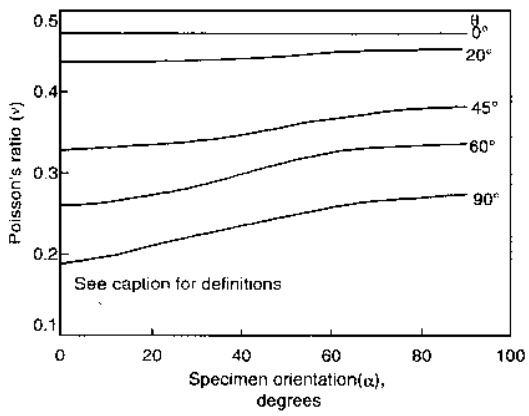
Elastic moduli for commercial purity titanium is higher than that of high-purity (99.9%) titanium, but interstitial impurities have less influence on temperature effects (H. Conrad *et al.*, *Titanium Science and Technology*, Plenum Press, 1973, p 970)
Source: Data from: P.E. Armstrong and H.L. Brown, *Trans. AIME*, Vol 230, 1964, p 962; and E.S. Fisher and C.J. Renken, *Phys. Rev. A.*, Vol 135, No. 2A, 1964, p 482

Variation of Young's modulus of single crystal titanium with orientation



Values were calculated from the compliance constants (S-constants). For example, when stress is applied parallel to the basal plane, then $E = 1/S = 100$ GPa (14.5×10^6 psi). For stress perpendicular to the basal plane, then $E = 1/S_{33} = 145$ GPa (21×10^6 psi).
Source: Reported in *Properties of Textured Titanium Alloys*, MCIC Report 74-20, Battelle, 1974, p 21

Theoretical variations of Poisson's ratio with orientation



Specimen orientation (α) is defined as the angle between the rolling direction and the specimen axis. The value of θ represents the angle between the basal pole and sheet normal.
Source: *Properties of Textured Titanium Alloys*, MCIC Report 74-20, Battelle, 1974, p 21

Shear Modulus

Poisson's Ratio. Like Young's modulus, the Poisson ratio of titanium depends on specimen orientation (see figure). Typical values for polycrystalline titanium range from 0.32 to 0.36.

Shear modulus (or modulus of rigidity) for pure titanium at room temperature ranges from about 42 to 45 GPa (6 to 6.5×10^6 psi) depending on the amount of interstitial impurities. In isotropic material, the shear modulus (G) is related to Young's modulus (E) as follows:

$$E = 2G(1 + \nu)$$

where ν is the poisson ratio. In textured material, this relation does not hold.

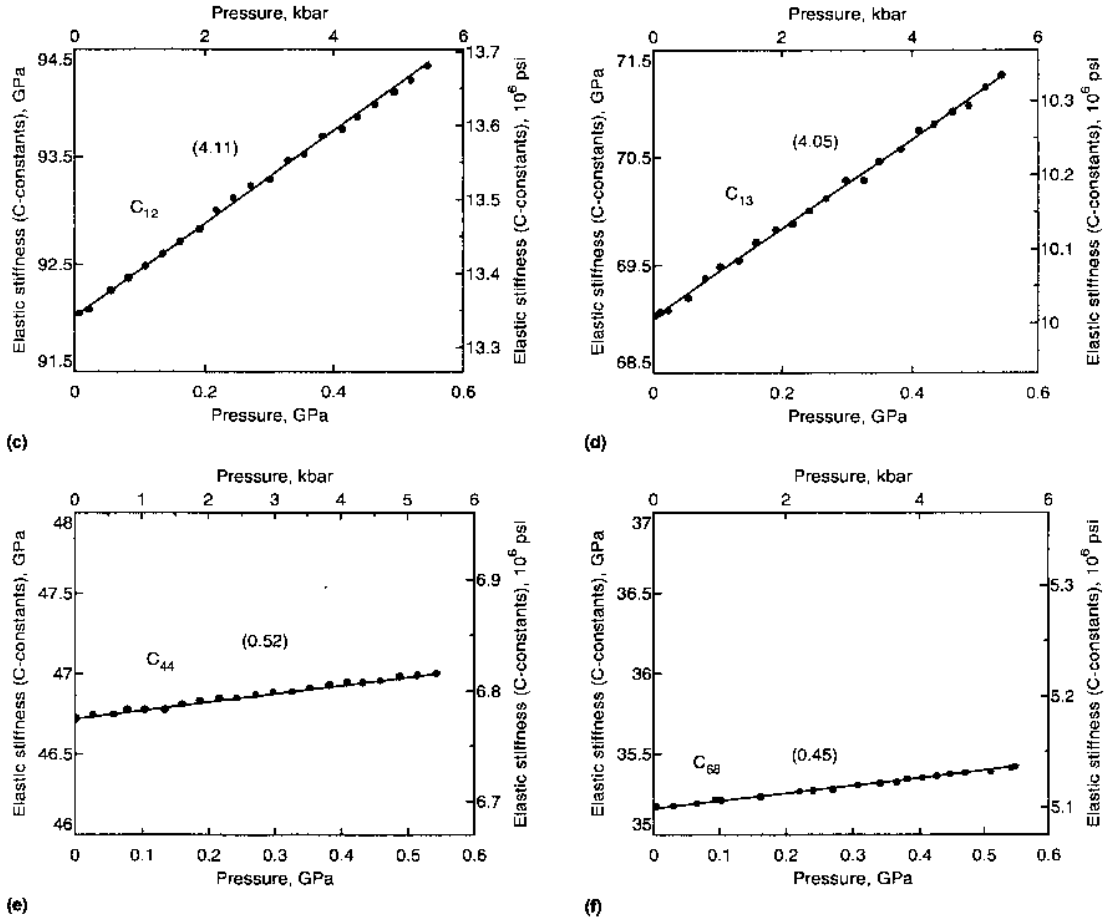
Bulk Modulus and Compressibility

Bulk Modulus. Bulk modulus (K) at room temperature is about 106 to 108 GPa (15.4 to 15.7 $\times 10^6$ psi) for pure titanium. The bulk modulus is related to Young's modulus as follows:

$$E = 3K(1 - 2\nu)$$

The bulk modulus also can be expressed as the reciprocal of compressibility in the elastic range. Therefore, a low bulk modulus of 90 GPa (13×10^6

Elastic stiffness moduli vs pressure of single crystal Ti (continued)



Numbers indicate values of pressure derivatives assuming a linear relation. Single crystals were obtained by zone melting of crystal bar iodide titanium..
Source: E.S. Fisher and M.H. Manghnani, Effects of Changes in Volume and c/a Ratio on the Pressure Derivatives of the Elastic Moduli of HCP Ti and Zr, *J. Phys. Chem. Solids.*, Vol 32, 1971, p 657-667

Young's Modulus

Typical values of Young's modulus (E) for pure titanium at room temperature is in the range of 100 to 110 GPa (15 to 16 × 10^6 psi). This range agrees with the evaluated elastic modulus of single titanium crystals in a direction perpendicular to the c-axis (see figure). However, a low elastic modulus of 85 GPa (12.3 × 10^6 psi) at 25 °C (298 K) is predicted from the following equation:

$$E \text{ (in GPa)} = 104.5 - 0.0645\,T$$

where T is the temperature in degrees kelvin (K). This equation is considered valid for high purity (99.9%) titanium below 500 K (W.H. Hill *et al.*, *Proc. ASTM*, Vol 61, 1961, p 890). For temperatures above 500 K (440 °F), the following equation is reported:

$$E \text{ (in GPa)} = 97 - 0.0486\,T$$

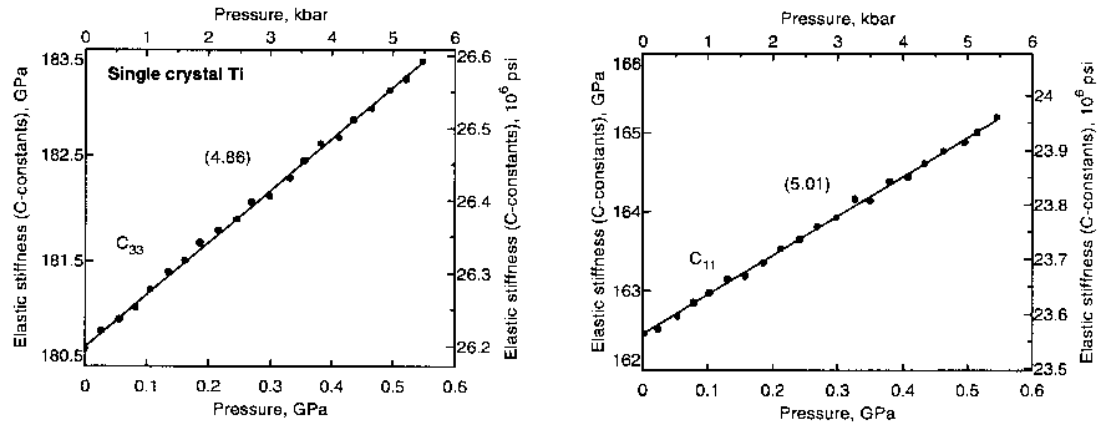
This equation for high purity titanium (99.9% Ti) predicts elastic moduli lower than that of commercial purity (<99.8%) titanium (see figure).

Single-crystal titanium: Effect of temperature on elastic stiffness moduli (*c*-constants)

Temperature		c_{11}		c_{33}		c_{44}		$c_{66} = 1/2(c_{11}-c_{12})$		c_{13}		c_{12}	
°C	K	GPa	10^{12} dyne/cm^2	GPa	10^{12} dyne/cm^2	GPa	10^{12} dyne/cm^2	GPA	10^{12} dyne/cm^2	GPa	10^{12} dyne/cm^2	GPa	10^{12} dyne/cm^2
-269	4	176.1	1.761	190.5	1.905	50.8	0.508	44.6	0.446	68.3	0.683	86.9	0.869
250	23	175.9	1.759	190.5	1.905	50.8	0.508	44.6	0.446	68.2	0.682	86.7	0.867
-200	73	174.9	1.749	189.4	1.894	50.5	0.505	43.9	0.439	68.0	0.680	87.1	0.871
-150	123	172.6	1.726	187.6	1.876	49.9	0.499	42.5	0.425	68.1	0.681	87.7	0.877
-100	173	169.9	1.699	185.7	1.857	49.0	0.490	40.5	0.405	68.4	0.684	88.9	0.889
-50	223	166.8	1.668	183.7	1.837	48.1	0.481	38.4	0.384	68.7	0.687	90.1	0.901
0	273	163.9	1.639	181.6	1.816	47.2	0.472	36.3	0.363	68.9	0.689	91.3	0.913
25	298	162.4	1.624	180.7	1.807	46.7	0.467	35.2	0.352	69.0	0.690	92.0	0.920
50	323	160.9	1.609	179.5	1.795	46.2	0.462	34.2	0.342	69.1	0.691	92.5	0.925
100	373	157.9	1.579	177.4	1.774	45.3	0.453	32.3	0.323	69.4	0.694	93.4	0.934
150	423	155.1	1.551	175.3	1.753	44.4	0.444	30.4	0.304	69.5	0.695	94.3	0.943
200	473	152.2	1.522	173.4	1.734	43.4	0.434	28.5	0.285	69.5	0.695	95.2	0.952
250	523	149.5	1.495	171.5	1.715	42.2	0.424	26.7	0.267	69.2	0.692	96.1	0.961
300	573	146.8	1.468	169.6	1.696	41.4	0.414	25.0	0.250	69.2	0.692	96.7	0.967
350	623	144.2	1.442	167.8	1.678	40.3	0.403	23.4	0.234	69.1	0.691	97.3	0.973
400	673	141.6	1.416	166.1	1.661	39.2	0.392	21.9	0.219	69.0	0.690	97.8	0978
450	723	139.2	1.392	164.4	1.644	38.1	0.381	20.5	0.205	69.2	0.692	98.3	0.983
500	773	136.8	1.368	162.7	1.627	37.0	0.370	19.1	0.191	68.8	0.688	98.5	0.985
550	823	134.5	1.345	161.0	1.610	35.9	0.359	17.8	0.178	68.8	0.688	98.8	0.988
600	873	132.2	1.322	159.3	1.593	34.8	0.348	16.6	0.166	68.8	0.688	99.1	0.991
650	923	129.9	1.299	157.6	1.576	33.7	0.337	15.4	0.154	68.8	0.688	99.2	0.992
700	973	127.6	1.276	156.0	1.560	32.6	0.326	14.2	0.142	...	...	99.3	0.993
750	1023	125.3	1.253	154.5	1.545	31.6	0.316	13.0	0.130	...	...	99.4	0.994
800	1073	123.1	1.231	152.9	1.529	30.7	0.307	11.8	0.118	...	...	99.6	0.996
810	1083	...	...	152.6	1.526	...	...	...	...	...	...	...	...
850	1123	121.0	1.210	...	...	29.7	0.297	10.7	0.107	...	...	99.6	0.996
880	1153	119.7	1.197	...	...	29.1	0.291	10.2	0.102	...	...	99.6	0.996
883	1156	...	...	150.4(a)	1.504(a)	29.1	0.291	10.0(a)	0.100(a)	68.8(a)	0.688(a)	99.6(a)	0.996(a)

(a) From extrapolated curve. Source: E.S. Fisher and J. Renken, *Physical Review*, Vol 135 (No. 2A), 1964, p A487

Elastic stiffness moduli vs pressure of single crystal Ti



(a) (b)

Numbers indicate values of pressure derivatives assuming a linear relation. Single crystals were obtained by zone melting of crystal bar iodide titanium.

Source: E.S. Fisher and M.H. Manghnani, Effects of Changes in Volume and c/a Ratio on the Pressure Derivatives of the Elastic Moduli of HCP Ti and Zr, *J. Phys. Chem. Solids.*, Vol 32, 1971, p 657-667

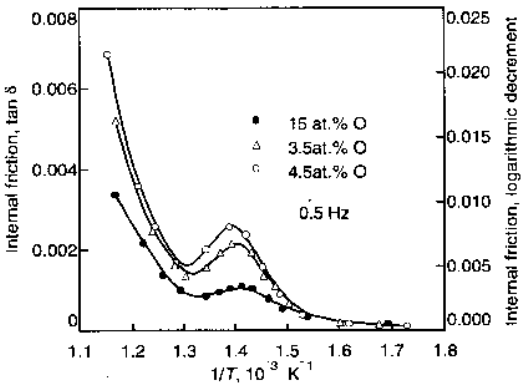
(continued)

Damping Properties

Internal Friction

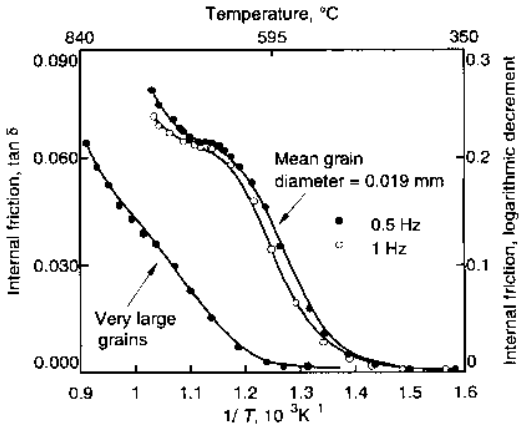
A Q^{-1} value of 1.5×10^{-4} for strain amplitudes of 10^{-6} to 10^{-3} is a reported estimate of pure titanium internal friction at an unspecified temperature. Relaxation spectra of pure titanium (see figure) reveal a small peak in stress relaxation at about 600 °C (1100 °F) for small-grained specimens. This peak is found at higher temperatures as oxygen levels increase. The introduction of oxygen also results in the appearance of a very small additional relaxation peak at about 450 °C (850 °F). At temperatures well above the 450 °C peak, internal friction increases as oxygen levels increase.

Effect of oxygen on internal friction



tan δ = logarithmic decrement/π.
Source: J.N. Pratt *et al.*, *Acta Metall.*, Vol 2, 1954., p 203-208

Pure titanium: Internal friction



tan δ = logarithmic decrement/π.
Small-grained specimens (0.019 mm mean diam) were tested at vibration frequencies of 0.5 Hz an 1 Hz, while the large-grained specimen was tested at 0.5 Hz.
Source: J.N. Pratt et al., *Acta Metall.*, Vol 2, 1954, p 203-208

Elastic Properties

Because of the α(hcp) structure of titanium, elastic properties depend on the orientation of the titanium crystals and the texture of polycrystalline titanium. Interstitial impurities (e.g. oxygen and nitrogen) and residual stress can also have a slight influence on elastic constants.

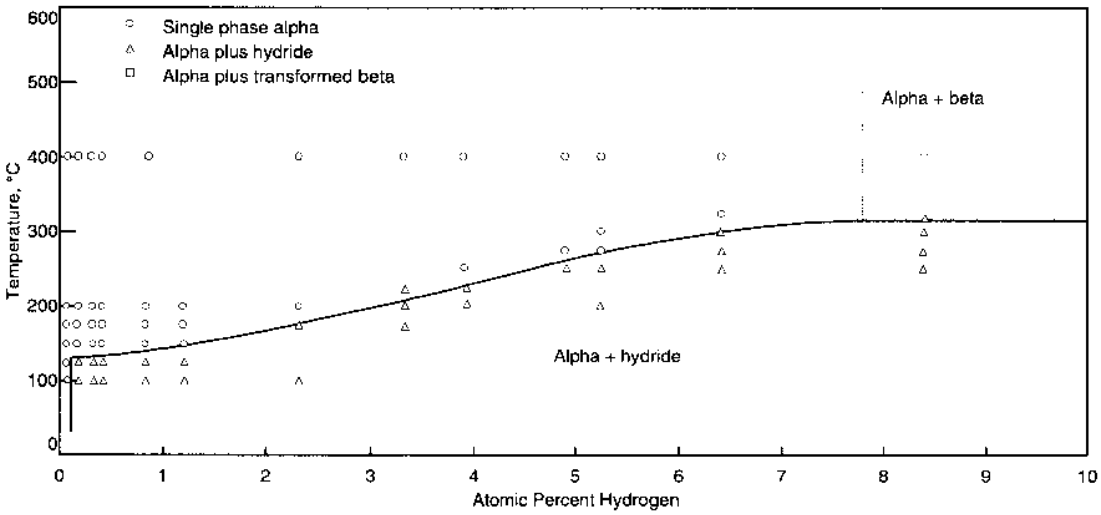
Elastic constants of HCP titanium single crystals

	Elastic stiffness (C constants)			Elastic compliances (S constants)	
C-Constant	10^6 psi	GPa	S-Constant	10^{-7} in.²/lbf	10^{-12}m²/N
C_{11}	22.33	153.97	S_{11}	0.688	9.98
C_{12}	12.47	85.98	S_{12}	–0.325	–4.71
C_{13}	9.74	67.16	S_{13}	–0.133	–1.93
C_{33}	26.54	183.0	S_{33}	0.476	6.90
C_{44}	6.71	46.27	S_{44}	1.475	21.4

Source: J.W. Flowers, Jr., K.C. O'Brien, and P.C. McEleney, "Elastic Constants of Alpha Titanium Single Crystals of 25 °C," *J. Less-Common Metals*, 1964, p 393-395, E.S. Fisher and D.J. Renken, "Single-Crystal Elastic Moduli and the HCP-BBC Transformations in Ti, Zr, and Hf," *Physical Review*, 135 (2A), 1964, p 482-494

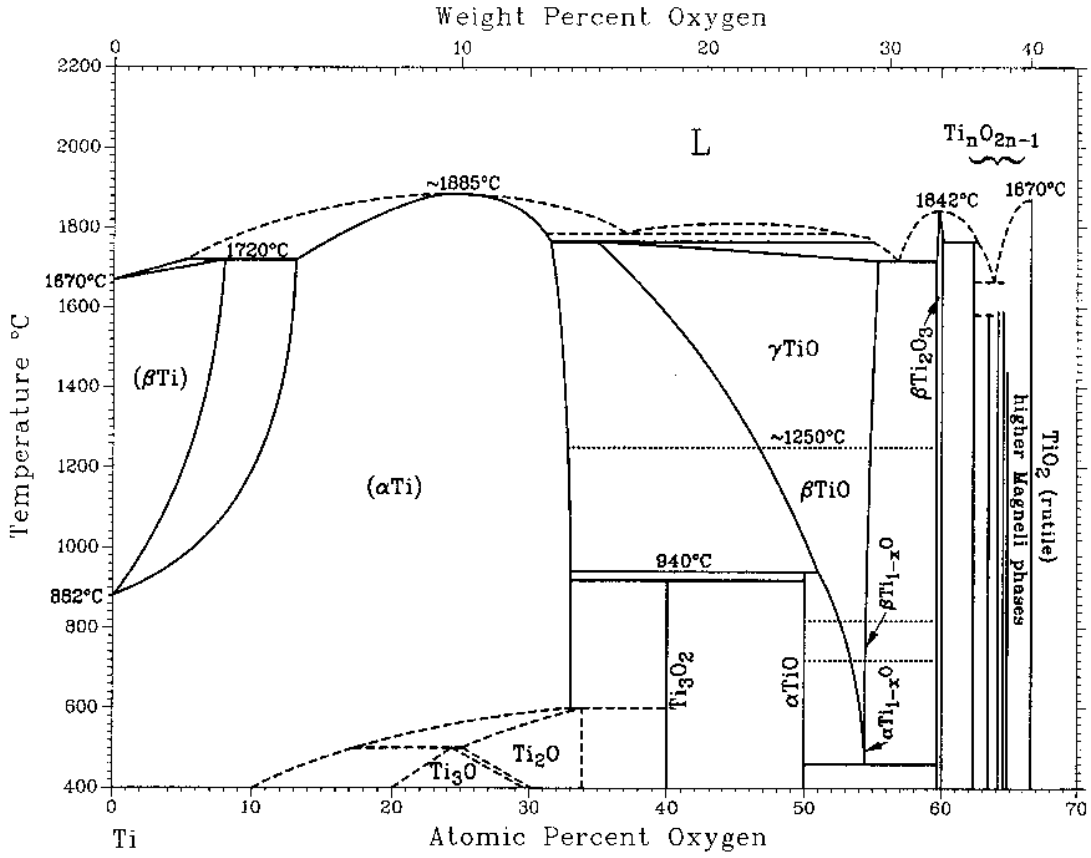
Selected Binary Phase Diagrams

Low temperature α solubility limit for hydrogen in high purity titanium



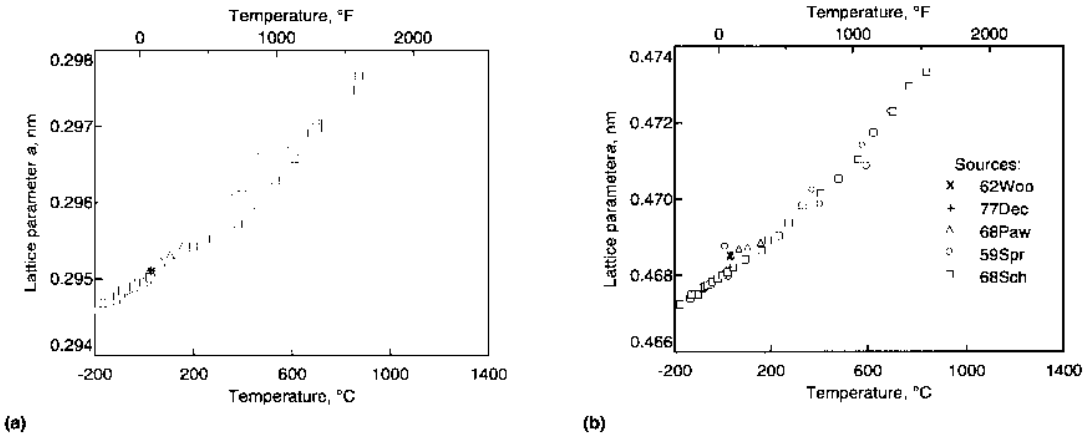
High-purity iodide titanium. The solubility of hydrogen in α titanium decreases from about 8 atomic pct at 300 °C to about 0.1 atomic pct at room temperature, the decrease in solubility being greatest down to 125 °C.
Source: G.A. Lenning, C.M. Craighead, and R.I. Jaffee, Constitution and Mechanical Properties of Titanium-Hydrogen Alloys, *Hydrogen Damage*, C.D. Beachem, Ed., American Society for Metals, 1977, p 100

Ti-O phase diagram



Source: J.L. Murray and H.A. Wriedt, 1987

Effect of temperature on lattice parameters

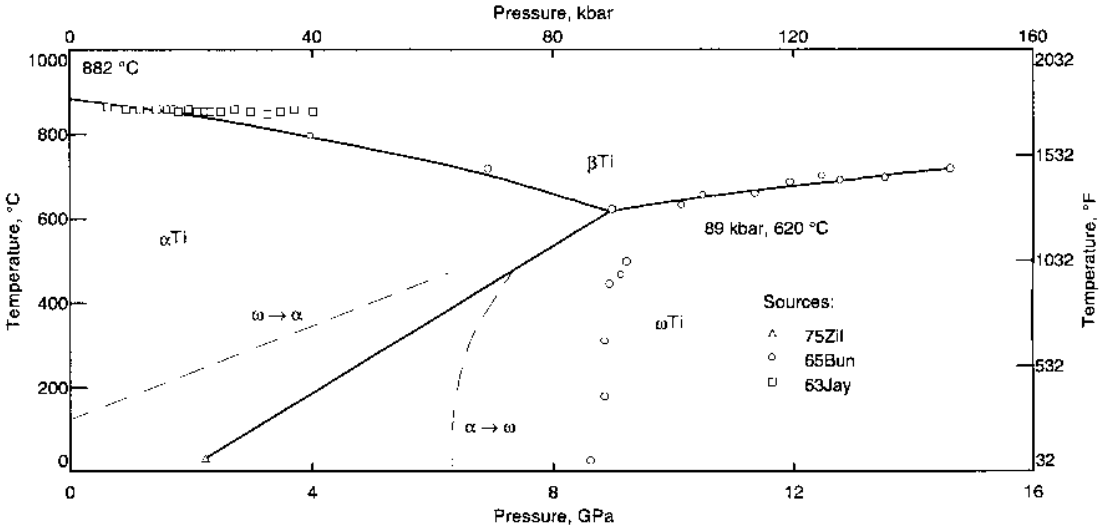


Source: *Phase Diagrams of Binary Titanium Alloys*, ASM International, 1987

Phase Diagrams

Pressure-Temperature Phase Diagram

Equilibrium crystal structure of pure titanium



Full citations of data sources are given in the figure reference.
Source: *Phase Diagrams of Binary Titanium Alloys*, ASM International, 1987, p 2

Lattice Parameters

Lattice Parameters of Pure α Titanium. Pure titanium has a *c:a* ratio of 1.587, which is significantly lower than that for other hexagonal metals such as magnesium, zinc, and cadmium. Lattice parameters are influenced by interstitial impurities (such as oxygen, principally) and temperature (see figures). Room-temperature lattice parameters (see table) include measurements on exceptionally pure titanium (90 to 4000 at.ppm O) by M. Dechamps *et al.*, *Scr. Metall.*, Vol 11, 1977, p 941-945 (77 Dec). Results were extrapolated to zero content and are accepted.

Lattice Parameter of Quenched ω Titanium. $a = 0.2813$ nm and $c = 0.4625$ nm.

Experimental determinations of the βTi ↔ αTi Transformation Temperature

Reference(a)	Transformation temperature (IPTS-68),°C	Experimental technique
[51Meq]	883	Hydrogen pressure
[51Duw]	882 ± 4	Cooling curve
[52Kot]	881	Drop calorimetry
[57Sco]	883 ± 2	Adiabatic calorimetry
[67Rud]	882	DTA
[74Cor]	~882	Electrical resistivity
[76Etc]	882 ± 2	Dilatometry
[78Cez](b)	893 ± 6	Pulse heating and resistivity
[84Mca]	883 ± 2	DTA (cooling, heating)
Assessed	882 ± 2	

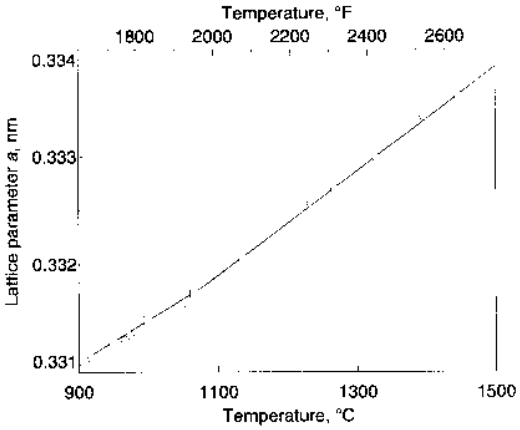
(a) Full citations given in *Phase Diagrams of Binary Titanium Alloys*, ASM International, 1987, p 1-4. (b) The transformation temperature value reported by [78Cez] is considered incorrect due to the use of pulse heating (2500 to 2700 °C/s).

Room-Temperature Lattice Parameters of αTi

Reference(a)	Lattice parameters, nm *a*	*c*	Temperature, °C
[49Cla]	0.29504	0.46833	25 ± 2
[49Gre]	0.29450	0.46845	25
[50Fin]	0.29504	0.46834	RT
[55Sza]	0.29506	0.46788	25
[59Spr]	0.29506	0.46797	22
[62Woo]	0.29511	0.46843	25 ± 2
[68Paw]	0.29508	0.46855	28
[68Sch]	0.29503	0.46810	20
[77Dec]	0.29512	0.46826	21 ± 1

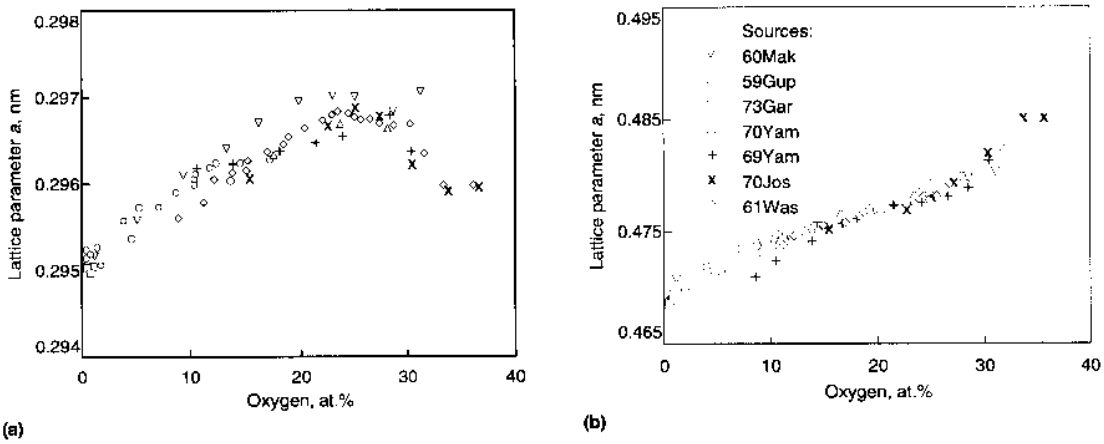
(a) Citations listed in *Phase Diagrams of Binary Titanium Alloys*, J.L. Murray, Ed., ASM International, 1987, p 1-5

Pure β titanium: Effect of temperature on *a*



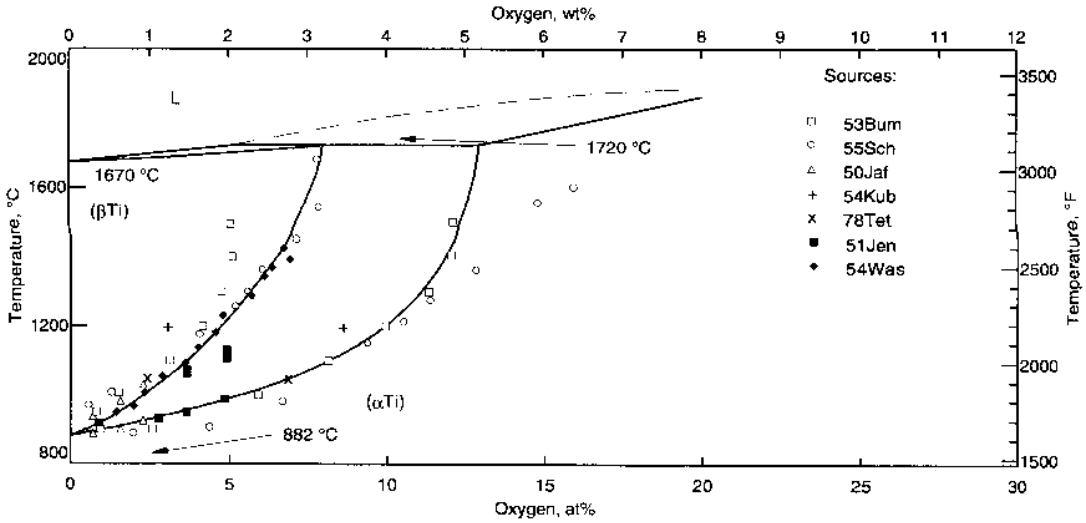
Source: J.L. Murray and H.A. Wriedt, Titanium, Phase Diagrams of Binary Titanium Alloys, J.L. Murray, Ed., ASM International, p 1-5

Pure α titanium: Effect of oxygen on lattice parameters



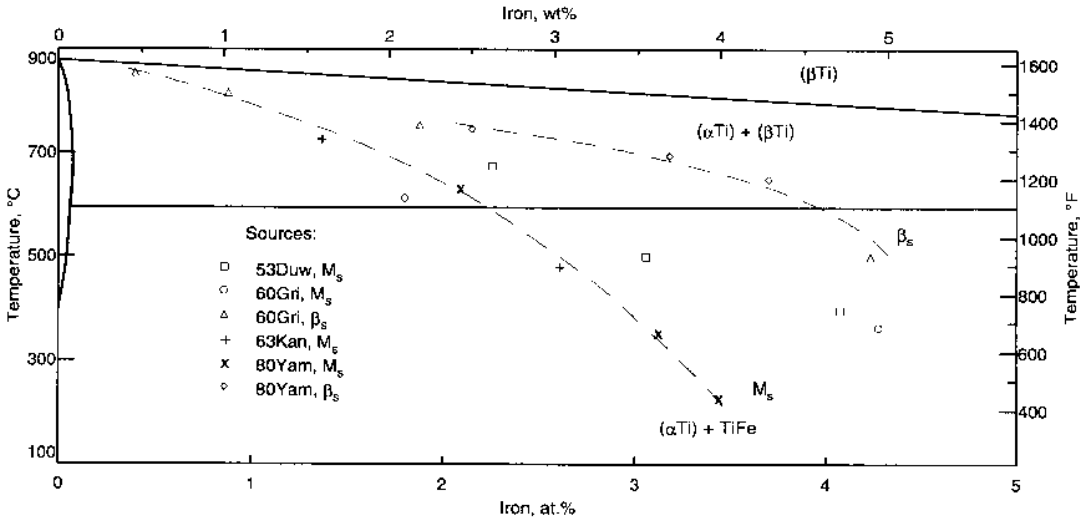
(a) Lattice parameter a. (b) Lattice parameter *c*
Source: Citations for the data sources are listed in *Phase Diagrams of Binary Titanium Alloys*, ASM International, 1987, p 211-229

Pure titanium: Effect of oxygen on α – β phase boundaries



See reference for full citations of data sources.
Source: J. Murray, Ed., *Phase Diagrams of Binary Titanium Alloys*, ASM International, 1987, p 214

Effect of iron on M_s and reversion (β_σ) temperatures



Full citations of data sources are given in the figure reference.
Source: J. Murray, Ed., *Phase Diagrams of Binary Titanium Alloys*, ASM International, 1987, p 104

are used, (αTi)/(βTi) phase boundary data are consistently found to extrapolate to 882 °C at pure Ti. The effect of O or N contamination is to raise the temperature of the transformation. Values as high as 900 °C have been reported in studies of binary systems, and the existence of an appreciable two phase αTi + βTi region in apparently pure Ti was due to contamination by O and/or N.

Martensite-Start Temperature. Above a critical cooling rate of 3000 °C/s (5400 °F/s), the (βTi) ↔ (αTi) transition is diffusionless (i.e., either martensitic or massive in mechanism). For unalloyed titanium, the martensite-start (M_s) temperature is influenced by oxygen and iron contents (see table and figure).

Martensite-start temperatures of Ti-O alloys

Composition, at.%	M_s temperature °C	°F
0	802 ± 10	1475 ± 18
0.6	836	
0.9	861	
1.0	901(a)	
1.2	849	
1.5	832	

(a) The maximum in M_s at about 1 at.% O was not explained.
Source: M. Cormier and F. Claisse, "Beta-Alpha Transformation in Ti and Ti-O Alloys," *J. Less-Common Met.*, Vol 34, 1974, p 181-189

Sonic velocities in pure titanium

Sample description	Velocity of sound, km/s (ft/s), for: Longitudinal (Compressive) waves	Transverse (Shear) waves	Surface waves	Reference
Unalloyed Ti (density unspecified)	~5(a) (16 390)	...	...	Fusfeld and Gilbert, *Phys. Rev.*, Series 2, Vol 77, 1950, p 302-303
Iodide titanium (crystal bar, 4.5 g/cm^3)	6.05 (19 850)	3.1 (10 170)	...	Fisher and Manghnani, *J. Phys. Chem. Solids*, Vol 32, 1971, p 657-667
Commercially pure titanium (4.5 g/cm^3)	6.10 (20 000)	3.12 (10 235)	2.79 (9150)	9th Edition *Metals Handbook*, Vol 17, p 235

(a) As reported by Lepkowski and Holladay (TML Report No. 73, Battelle, 1957) without specification of waveform type

Nuclear Properties

Thermal Neutron Cross Section. The absorption cross section for thermal neutrons is about 5.6 ± 0.4 barns (0.070 cm^2/g) for the normally occurring isotopic mixture of metallic titanium. Scattering cross sections for natural isotopes are:

Stable isotopes of titanium

Isotope	Natural abundance, %	Cross section, b(a)
^{46}Ti	7.95	0.6
^{47}Ti	7.75	1.6
^{48}Ti	73.43	8.0
^{49}Ti	5.51	1.8
^{50}Ti	5.34	0.2

(a) b, barns. Source: *The Reactor Handbook*, Vol I, Atomic Energy Commission, AECD 3645, 1955, p 352

Surface Properties

Surface Tension. 1.2 N/m (1200 dynes/cm)

Work Function. The work function is the energy necessary to remove an electron from the surface and is high if the cohesive energy is high. The work function is also affected by oxidation or other surface layers. Reported values range from 3.95 ± 0.02 eV to 4.45 ± 0.05 eV (W. Lepkowski and J. Holladay, TML Report No. 73, Battelle, 1957).

Total Hemispherical Emittance. 0.30 at 710 °C

Optical constants and reflectance of titanium

Wavelength, μm	Refractive index (n)	Absorption coefficient (k)	Reflectance (R) calculated,% (a)	Reflectance (R) measured,% (a)
436	2.04	2.85	53.0	51.3
546	2.53	3.33	57.0	55.8
578	2.64	3.42	57.0	56.5
650	3.03	3.65	59.0	58.5

(a) R = reflectance for normal incidence. Source: W.J. Lepkowski and J.W. Holladay, TML Report No. 73, Battelle, 1957, p 67

Thermal Properties

More detailed assessments of the following thermal properties are given in subsequent sections of this datasheet.

Melting Temperature. 1670 ± 5 °C

Boiling Point. 3260 °C (estimated)

Vapor Pressure. From 1587 to 1698 K:

$$\log P = 7.7960 - \frac{24\,644}{T} - 0.000227\,T$$

where P is in Pa and T is in K

Phase Transformation Temperature. α to β, 882 ± 2 °C (1620 °F)

Coefficient of Thermal Expansion. At 20 °C, 8.41 × 10^{-6}/°C

Specific Heat. 523 J/kg · °C at room temperature

Latent Heat of Fusion. ~292 kJ/kg

Latent Heat of Transformation. ~85 kJ/kg (estimated)

Latent Heat of Vaporization. 9.83 MJ/kg (estimated)

Thermal Conductivity. ~17 W/m · °C at room temperature (see the section "Transport Properties" for a more detailed assessment).

Phases and Structures

Transition Temperatures

Beta Transus. At atmospheric pressure, titanium has the α(cph) structure at low temperature and transforms at 882 ± 2 °C (1620 ± 3.5 °F) to the high-temperature β(bcc) structure. The assessed transformation temperature of 882 ± 2 °C is based on phase diagram data for binary Ti systems (see figure for Ti-O system) as well as on direct measurements (see table). When alloys of high purity